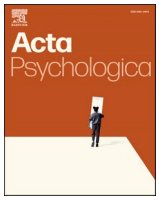
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What are the intellectual foundations and thematic structures shaping flow experience research in tourism?

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ABSTRACT

This study addresses the fragmented state of research on flow experience in the tourism sector, an increasingly important concept in the modern experience economy. The main objective was to systematize and map the intellectual structure, development, and future trajectory of this field. Applying a hybrid bibliometric-systematic literature review method, the paper analyzed 118 articles from the Scopus database (2011–2025) using VOS-viewer and Bibliometrix. The study's conclusions were further reinforced and validated through a comparative analysis with data extracted from the Web of Science database. Key findings revealed two distinct research phases, with a significant boom after 2020 driven by the impact of the COVID-19 pandemic and the rise of virtual reality (VR) tourism. Thematic analysis revealed a clear knowledge structure, with basic themes serving as the theoretical basis, while motor themes (authenticity and tourism live streaming) drove the current research. This landscape was divided into a psychological core, focused on exploring the nature of the experience, and a managerial core, centered on applying the flow experience for strategic goals. This study contributes a new conceptual framework that organizes the research field into antecedents, flow state, and outcomes, while also providing practical guidance for managers to design engaging experiences.

1. Introduction

The 21st century has witnessed a distinct shift away from a service-driven economy toward what many now refer to as the experience economy. Tourism, perhaps more than any other sector, illustrates this change in society. What matters is not only the provision of reliable services but also, and increasingly, the crafting of experiences that may be distinctive, memorable, and, at times, even disruptive. The same logic extends beyond tourism to fields such as culture, retail, entertainment, and food, where the service is essentially the experience (De Matos et al., 2020). This shift also has geographical consequences. Urban tourism, for instance, appears to be increasingly centered on both experience production and consumption (Chang & Huang, 2014). Today, travelers do not simply purchase services. Many people seek encounters that stir their senses, move their emotions, and leave lasting memories. It is not surprising, then, that destinations frame their promotions by emphasizing how visitors might feel, think, act, or connect with others (Ketter, 2018; Schlesinger et al., 2020). Tourism research usually groups such experiences into four categories: entertainment, aesthetics, escape, and education (Luo et al., 2018; Sipe & Testa, 2018). When combined, these dimensions create a more comprehensive and persuasive experience for travelers. Notably, experiences remembered most strongly are also those that shape satisfaction, strengthen emotional commitment, and encourage visitors to share their stories with others (Guleria et al., 2024; Sihombing & Antonio, 2025).

This economic transformation has created an urgent demand in academia for theoretical frameworks that are strong enough to decode and analyze such deep states of interaction. For managers and researchers, the question is not only what makes a good experience but also why and how certain encounters can become engaging, memorable, and valuable. Customer experience is, nonetheless, complex and multi-dimensional with sensory, emotional, cognitive, behavioral, and social aspects (Sidaoui et al., 2020). Through this perspective, flow has emerged as a central idea in understanding the way in which travelers gain optimal experiences, achieve satisfaction, and develop lasting memories. The increasing popularity of flow among tourism scholars is part of a larger move toward experience-oriented models, in which interactive quality and emotional engagement are viewed as primary

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drivers of tourist activity and destination performance (deMatos et al., 2021; Karasakal & Albayrak, 2022). Flow, which originated in positive psychology, has a particular application in the tourism domain because of its value in enhancing satisfaction and participation. It is generally characterized by intense focus, loss of self-awareness, and distorted sense of time, all of which are objective states of experience that culminate into stimulating and pleasurable experiences (Armağan & Temel, 2022; Lin et al., 2024). Such states never arise autonomously. They are influenced by a range of factors, including destination attributes, novelty, and interpersonal interactions (deMatos et al., 2024).

As a domain primarily focused on experiential experiences, tourism offers a fertile, organic setting for the study and application of the flow concept. It seems that by providing appropriate conditions for flow, tourism enterprises can positively influence visitor experiences in terms of engagement, satisfaction, and even well-being, thereby making experiences more significant and impactful (Armağan & Temel, 2022; Fansurya et al., 2024). A considerable body of research supports models in which specific antecedents, flow states, and outcomes are connected within relatively specific domains. For example, studies on tourism websites and online communities examine website design elements and system characteristics as flow stimuli, and the induced flow, in turn, relates to satisfaction (Gao et al., 2017; Gao & Bai, 2014). Other studies investigate the relationships between perceived value and flow, and subsequent relationships between flow and outcomes such as loyalty, environmental behavior, and memorable experiences in nature and leisure environments (deMatos et al., 2024; Fansurya et al., 2024; Kim & Thapa, 2018). While the models and studies are highly valuable at the micro-level, the overall literature, however, fails to offer a clear and higher-level synthesis that effectively brings together the fragmented relationships between the variables and presents a unifying antecedents-flow-outcomes approach for the field as a whole, thereby facilitating the development of a structured research agenda by integrating the contributions of the studies to the overall approach.

While research on flow experiences in tourism has developed rapidly, the literature remains fragmented in some respects. The literature has not provided a comprehensive, field-wide account of the development of research on flow experiences, including publication volume, turning points, and post-COVID developments, particularly in relation to technology-mediated experiences (Kim & Hall, 2019). Detailed reviews are helpful in this sense, as they establish the theoretical foundations of tourism experiences, such as flow, and provide greater insight into tourist attitudes and behavior (Chen, 2025; Xu et al., 2022). They also enable researchers and practitioners to understand better how travelers perceive and engage with activities across various experiential contexts (deMatos et al., 2021).

However, most of the existing conceptual reviews and contributions tend to focus on specific topics related to the experience—such as customer experience in tourism (Adhikari & Bhattacharya, 2016), memorable tourism experiences (Hosseini et al., 2023), or VR-based destination experiences (Huang et al., 2020), rather than offering a bibliometric mapping of the core authors, journals, and thematic clusters that collectively structure flow research in tourism. Therefore, the psychological foundation of the flow theory, including the mechanisms, measurement, and experiences, as well as the managerial implications of the theory, such as satisfaction, loyalty, technology, and well-being, are often studied separately rather than under a unified knowledge framework (Buckley et al., 2022). This prompts an exploration of the key actors and thematic structures underlying this field.

To address these gaps, a bibliometric analysis combined with a systematic literature review (B-SLR) has been proposed as a rigorous, objective tool. This hybrid approach leverages the best features of both methods to make field mapping faster and more efficient (Marzi et al., 2025; Pulsiri & Vatananan-Thesenvitz, 2018). The B-SLR method involves several steps to evaluate contributions based on how well they reflect the topic, their timeliness, coverage, rigor, coherence, and novelty. In this way, it provides a solid basis for advancing research in tourism (de Bruyn et al., 2023; Panigrahy & Verma, 2024). Based on this approach, the current study systematically evaluates and analyzes existing literature on flow experiences in tourism based on the following four research questions (RQs):

(RQ1) How has the number of articles published on flow experience in tourism evolved over time?
(RQ2) Which journals, articles, authors, and geographical regions are the most influential in this field?
(RQ3) What are the main themes and key impacts of each thematic group within the research field?
(RQ4) What future research directions will be pursued in the field of flow experience in tourism?

2. Literature review

2.1. Flow theory: conceptual foundations

Flow theory, created by Mihaly Csikszentmihalyi, involves being completely engrossed in an activity, which is termed the optimal experience or the flow state (Biasutti, 2011). In this situation, a person is able to concentrate intensely, achieve a level of mastery, become oblivious to self, and experience a distortion of time (Palomäki et al., 2021). The flow experience is characterized by total attention, a balance of challenge and skill, distinct tasks with immediate results, self-absorbed awareness, and time distortion (Fullagar & Fave, 2017).

Csikszentmihalyi (1990) and Jackson and Marsh (1996) developed the flow theory into nine core aspects detailed in Table 1. These nine aspects include (1) challenge–skill balance: perceived challenges exceed one's skill, and (2) goal clarity: goals that provide direction for action, (3) the unambiguous feedback: feedback that tells one how well they are doing, (4) action–awareness merging: behavior is more involuntary and automatic, (5) concentration on the task: the total focus on the task, (6) sense of control: the self- feeling that they can control the outcomes, (7) loss of self-consciousness: the self and social concern, (8) time transformation: feeling that time is moving faster than it actually is, and (9) autotelic experience: the activity is internally pleasurable and is done for its own sake.

Table 1 provides a synthetic overview of the construct's intellectual foundations. It compares flow with six commonly used concepts in tourism and media research: mindfulness, engagement, satisfaction, presence, immersion, and peak experience. For each construct, the table summarizes key theoretical roots, core definition, experiential characteristics, typical measurement instruments, and its typical role within tourism models (driver, mediator, outcome). By organising these elements side-by-side, the table clarifies that flow is not a generic positive feeling but a structured, goal-directed state centred on a balance of challenge and skill, clear goals, and intrinsic enjoyment (Fullagar & Fave, 2017; Jackson & Eklund, 2002).

2.2. Psychological mechanisms underlying flow experience

The primary role of Table 1 is to address the flow construct in relation to other related constructs in tourism and media studies. Although these constructs can occur simultaneously within a single experience, the table highlights some of the most salient distinctions among concepts and functions. To begin with, flow is distinguished from mindfulness and engagement. Mindfulness is being aware of the present moment without judgment, typically without specific performance/achievement goals. At the same time, engagement is a sustained state of vigor and, in some cases, an absorbed, dedicated state toward the activity. Flow is described as a brief, peak experience characterized by the right amount of challenge and a defined performance framework (Bakker, 2008; deMatos et al., 2021). A peak experience is a rare, sometimes transcendent experience that can have spiritual or existential significance (Knobloch et al., 2016). Peak experiences can include some

Table 1
Distinguishing flow from related psychological constructs.

Construct	Core definition	Dimensionality	Distinguishing features	Tourism application	Measurement approach
Flow (flow state, flow experience)	Optimal absorption state with challenge-skill balance	9 dimensions: challenge-skill balance, clear goals, feedback, merged action-awareness, concentration, control, loss of self-consciousness, time distortion, autotelic experience	- Requires optimal challenge-skill ratio - Intense, time-bounded state - Intrinsically rewarding - Action-focused	Adventure tourism, cultural immersion, VR experiences	Flow State Scale-2 (Jackson & Eklund, 2002); psychophysiological measures (Huang et al., 2020)
Mindfulness	Non-judgmental present-moment awareness	5 facets: observing, describing, acting with awareness, non-judging, non-reactivity	- Observation without goal - Sustained state - Acceptance-focused - No challenge requirement	Wellness tourism, meditation retreats, contemplative travel	Five Facet Mindfulness Questionnaire (Baer et al., 2008)
Engagement	Persistent positive activity-related state	3 dimensions: vigor, dedication, absorption	- Can be sustained long-term - Moderate absorption - Does not require peak challenge - Extrinsic rewards acceptable	Loyalty programs, destination activities, service encounters	Utrecht Work Engagement Scale (Schaufeli & Salanova, 2006)
Satisfaction	Post-experience evaluative judgment	Cognitive and affective components	- Retrospective evaluation - Expectation-based - Cognitive judgment - Outcome-focused	Overall trip evaluation, service quality assessment	SERVQUAL (Parasuraman et al., 1991), Expectation-Confirmation Theory (Oliver, 1977)
Presence (Telepresence)	Subjective sense of "being there"	3 dimensions: spatial, social, self-presence	- Technology-mediated - Perceptual illusion - No intrinsic motivation requirement - Can be passive	VR tourism, live-streaming, metaverse travel	ITC-Sense of Presence Inventory (Lessiter et al., 2001)
Immersion	Objective sensory engulfment	Continuum (low to high)	- Technology characteristics - Sensory-focused - Externally defined - No skill component	Multi-sensory experiences, themed attractions, VR	System Immersion Questionnaire (Jennett et al., 2008)
Peak Experience	Rare transcendent moments	Holistic, ineffable	- Rarer than flow - Often spiritual/ transcendent - Less predictable - No challenge requirement	Destination epiphanies, transformative travel	Peak Experience Scale (Privette, 1983)

features of flow, and they are not easily engineered or reliably measured with traditional psychological tests.

On the one hand, satisfaction represents a post hoc evaluation, in which expectations are juxtaposed with perceived performance; therefore, satisfaction is measured after the experience and is often used as a primary outcome in tourism-related models (Fournier & Mick, 1999; Giese & Cote, 2000). Flow, on the other hand, represents an in situ experiential state, often being used as a mediator in stimulus-organism-response (SOR) sequences, connecting stimuli such as attributes of a destination or VR design with outcomes like satisfaction, loyalty, or word-of-mouth, respectively (Wu & Liang, 2011).

The feeling of presence represents a sense of being there in a virtual environment, whereas immersion is mainly a function of the sensory richness and interactivity of the system, with less emphasis on presence (Biocca et al., 2003; Hudson et al., 2019). Technology has a significant impact on presence. In contrast, flow includes psychological components such as challenge–skill balance, perceived control, and intrinsic pleasure. Thus, presence and immersion are primarily presented in Table 1 as technological or perceptual attributes, whereas flow is a higher-order construct that may, but does not necessarily, result from high levels of presence and immersion (Huang et al., 2020; Kim et al., 2020).

2.3. Flow in tourism contexts

As a concept rooted in positive psychology, the flow experience holds particular importance in tourism, as it has the potential to enhance visitor satisfaction and create memorable experiences. It is usually defined by heightened concentration, reduced sense of self, and distorted time estimation, which all contribute to creating profound absorption and enjoyment in a task (Armağan & Temel, 2022; Elliot, 2021). More in-depth descriptions are when the tourists are completely immersed in activities such as adventure activities or cultural engagement, where flow states are more likely attained, and overall satisfaction and active recall of the experience are greater (Karasakal & Albayrak, 2022; Wang et al., 2023). In such states, travelers lose track of time and become completely absorbed in the activity, thereby increasing the sense of enjoyment and completeness (Lu et al., 2022). Flow, however,

tends to produce positive outcomes primarily when it is driven by intrinsic motivation—when individuals engage in activities for the enjoyment and satisfaction they provide rather than for external rewards.

Flow experiences are typically associated with adventure tourism activities such as mountain climbing, white-water rafting, or other outdoor activities where skill and challenge are carefully negotiated (Giddy, 2020; Wu & Liang, 2011). Such activities generate high levels of participant engagement and satisfaction. At the same time, flow also plays an important role in cultural and heritage tourism. For instance, visitors to historic sites such as ancient cities may experience flow through cognitive engagement, which can strengthen both satisfaction and loyalty (Aykol et al., 2025). Cultural tourists, therefore, may display different flow states depending on their motivation and level of interaction (Wang et al., 2023).

3. Materials and methods

3.1. Data collection

This study is based on articles produced by Elsevier's Scopus database. The choice of Scopus over other top databases, such as Web of Science (WoS), was a deliberate strategic choice given the specific orientation and trajectory of development of flow experience studies in tourism studies. Although both databases are reputable, their coverage and focus differ in ways that are important for the goals of this study. Scopus has broader journal coverage compared to WoS, particularly in the social sciences, arts, and humanities. This is because Scopus contains a more comprehensive collection of international and emerging journals (Alryalat et al., 2019; Asubiaro et al., 2024). Scopus also covers local and regional journals more, hence they have a broader international coverage. This is particularly beneficial for interdisciplinary studies such as tourism, which integrates research from other fields (Budimir et al., 2021).

In order to reduce any possible database selection bias, an identical research strategy was subsequently carried out in Web of Science (WoS), with results being compared with those in Scopus, in line with previous research indicating that citation-based indicators calculated independently in each of these two databases are almost perfectly correlated at the field level (Archambault et al., 2009). Instead of combining results mechanically from WoS and Scopus, which recent methodological research has shown can result in considerable inconsistencies and artefacts in citation networks (Kumpulainen & Seppänen, 2022), this current study, in line with recent bibliometric best practices, carries out all performance and science-mapping analysis on the Scopus dataset alone, with WoS being used as an external validation dataset.

Data was collected and queried up to June 12, 2025, by performing unique complex searches in the Title-Abstract-Keyword fields of the Scopus database. The exact query string is provided in Table 2. This query chain is systematically designed to maximize both coverage and accuracy through a multi-layered logical structure:

(1) The concept in the mind is a flow with the first search field ("flow experience" OR… "optimal experience") aimed at comprehensively capturing the core psychological structure by using the most common synonyms, ensuring a broad network is established for relevant theoretical discussions.
(2) The concept of tourism context is mentioned in the second proposition (touris* OR travel*… "visitor experience") to identify the field of application in the study. The use of the wildcard character "*" and a range of broad terms ensured that studies from different sub-sectors (such as hospitality, recreation, and virtual tourism) were included.
(3) The origin of the theory with the author search field ("csikszentmihalyi m*") is a vital methodological component. This is not a simple word search; it is a theoretically grounded search.

Table 2
Data acquisition. Source: compiled by the author.

Description	Conditions	Number of Documents
Search query	TITLE-ABS-KEY (("flow experience" OR "flow state" OR "flow theory" OR "optimal experience") AND (touris* OR travel* OR hospitality OR leisure OR recreation* OR "virtual reality" OR "visitor experience")) AND (REF ("csikszentmihalyi m*")) AND (LIMIT-TO (DOCTYPE, "ar") AND (LIMIT-TO (LANGUAGE, "English")) AND NOT TITLE-ABS-KEY ("water flow" OR "cash flow" OR "optic flow" OR "blood flow" OR "data flow")	n = 395
Query search date	June 12, 2025	
Database	Scopus	
Research areas screened according to PRISMA	Expansion and development in the field of flow in tourism cluster. Flow theories/ experience/state in Tourism (including virtual reality tourism)	n = 181
Type of article	Journal article	
Years	All year (2011–2025)	
Language	English	

(4) Exclusion criteria are added to the NOT TITLE-ABS-KEY clause based on keywords ("water flow" OR "cash flow" OR…) that act as a filter to exclude publications containing the term "flow". However, they are not relevant to the research topic.
(5) Finally, the overall search filter was limited to DOCTYPE "ar" (journal article) to focus on original, peer-reviewed research and the language "English" to ensure a linguistically coherent dataset for analysis. The timeframe was set from 2011 to 2025.

The selection process adhered to the four-stage flow diagram recommended by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement to guarantee transparency and reproducibility. The first search returned 395 documents, which were then carefully screened to get a final set of 118 articles for analysis:

(1) Stage 1: Identification (n = 395): The initial query in Scopus identified 395 publications. At this stage, automated tools within the directory management software were used to identify and remove any duplicate records.
(2) Stage 2: Screening (n = 395 → n = 181 → n = 145): This stage involved a two-step screening process. The initial 395 publication list was manually filtered at the database level by "research field", which excluded 214 records that appeared in research fields not included in the tourism, hospitality, and social science domain, resulting in a pool of 181 potentially relevant articles. Next, title and abstract screening of 181 articles was conducted, followed by a manual review by two independent researchers. Articles were excluded at this stage if they clearly did not meet the criteria, including (such as the context not being related to tourism despite having keywords, the study being a review or book review rather than an experimental article, or the concept of "flow" being used metaphorically rather than as a central theoretical construct. This screening excluded 36 articles, resulting in 145 remaining articles.
(3) Stage 3: Eligibility (n = 145 → n = 118): The full texts of the remaining 145 articles were retrieved and assessed for eligibility. Articles were excluded at this final stage for reasons such as a lack of precise theoretical application of flow theory, methodological limitations that could affect their validity for inclusion, or a focus on a peripheral aspect of tourism without a direct link to the tourist experience. A total of 27 studies were excluded during the full-text review process.

(4) Stage 4: Includes (n = 118): The final dataset included 118 journal articles that met all inclusion criteria. These articles formed the basis for all subsequent analyses.

3.2. Data preprocessing and analyses

The final dataset used in the analysis included 118 publications with complete information extracted from Scopus (specifically, full information on title, authors, affiliation, abstract, keywords, and references is required), standardized into a file that can be run on both VOSviewer and Bibliometrix. First, the data were analyzed using Bibliometrix, a bibliometric software package in R that is renowned for its capability to map scientific literature for bibliometric analysis (Aria & Cuccurullo, 2017). Subsequently, VOSviewer was used to convert these relationships into visual formats, including network visualizations of keyword co-occurrence and author linkages (van Eck & Waltman, 2014). This combination was chosen because each tool has its own strengths, which makes for a strong and repeatable analytical workflow.

Bibliometrix in RStudio supports science mapping techniques, including keyword co-occurrence, co-authorship, and co-citation analyses. Numerous of these techniques aid in the visualization of conceptual, social, and intellectual bases of researchers' topics (Chansanam & Li, 2023; Song et al., 2021). Importantly, Bibliometrix contains strong visualization capabilities, via which researchers are able to prepare graphical representations of bibliometric data. These include popular topics, bibliographic coupling, and co-occurrence networks, which are central to detecting up-rising fields of study and dynamics of scientific knowledge (Kilicoglu & Mehmetcik, 2021; Mazlee & Zunairah, 2024). VOSviewer, on the other hand, is a specialized computer program that can easily visualize bibliometric networks, such as citations, bibliographic coupling, co-citations, and co-authorship relations (Li & Wei, 2022; Van Eck & Waltman, 2009). VOSviewer places a strong emphasis on the graphical representation of bibliometric maps, making it particularly useful for displaying large bibliometric maps in a clear and intelligible manner (van Eck & Waltman, 2010).

Table 3 summarizes the entire analysis process. It is divided into two parts that work together: (1) performance analysis, which attempts to measure the impact and productivity of research actors, and (2) science mapping, which attempts to show the knowledge structure and evolution of the research field. Each technique was selected to address the specific research questions (RQs) outlined in the Introduction section.

Table 3
Overview of applied bibliometric analyses. Source: compiled by the author.

Analysis technique	Research question	Purpose & key metrics	Software
Performance analysis			
Annual Scientific Production	RQ1	Track the field's growth; Annual Growth Rate (AGR)	Bibliometrix (R)
Most Relevant Authors	RQ2	Identify influential researchers (Lotka's Law); Total Citations (TC), TC per Year (TCpY)	Bibliometrix (R)
Most Relevant Sources	RQ2	Identify core journals (Bradford's Law); Number of articles, Local H-index	Bibliometrix (R)
Most-Cited Documents	RQ2, RQ3	Identify foundational works; Global Total Citations, TCpY	Bibliometrix (R)
Science mapping			
Thematic Map Analysis	RQ3, RQ4	Map the conceptual structure; Centrality & Density (Callon's)	Bibliometrix (R)
Keyword Co-occurrence Network Analysis	RQ3	Identify research theme clusters; Co-occurrence frequency	VOSviewer
Author Co-citation Network Analysis	RQ2	Outline knowledge bases and schools of thought; Co-citation frequency	VOSviewer
Trend Topics Analysis	RQ4	Identify emerging research trends; Keyword frequency over time	Bibliometrix (R)

4. Results

4.1. Annual scientific production

Analysis of annual scientific production data on the topic of "flow experience in tourism" from 2011 to 2025 provides a clear picture of the field's development. The data, which include 118 articles published over 15 years, reflect not only an increase in quantity but also a fundamental paradigm shift in research priorities in tourism studies, especially in the post-COVID-19 period.

Table 4 clearly shows that the number of publications changed significantly between 2011 and 2025. The annual growth rate (AGR) was volatile, especially before 2021, when there were some years of negative growth (−33% in 2015, −67% in 2020). However, after 2020, there was an unprecedented rise in publications. The AGR even hit triple digits for three consecutive years (250% in 2021, 114% in 2022, and 100% in 2023). The 3-year moving average further confirms this trend, shifting from a relatively flat line (around 2–5 articles per year) between 2013 and 2020 to a steep upward curve, peaking at 23.0 articles per year during the 2022–2024 period. Based on these findings, we can clearly divide the research history on flow experience in tourism into two distinct phases:

The Latent & Exploratory Phase (2011−2020): This period, which lasted a full decade, was marked by low and inconsistent publication numbers, swinging between zero and six articles per year. In total, only 27 articles were published during those 10 years, making up a mere 22.9% of all research done over the entire 15-year period. The annual growth rate (AGR) varied significantly, with some years showing no growth (0%) or even experiencing sharp declines. This suggests that "flow experience in tourism" was likely a niche topic of interest to only a small group of researchers.

Exploratory studies into adventure and physical tourism experiences highlighted this phase. Two pioneers, Wu and Liang (2011), applied flow theory to adventure tourism, specifically whitewater rafting. They found that skills, challenges, and playfulness were important predictors of the flow experience, which in turn led to tourist satisfaction. Cheng

Table 4
Publication productivity and growth rate of flow experience in tourism. Source: author's synthesis from analysis results on Bibliometrix.

Year	Articles	Percentage of no. of articles published per year (%)	Cumulative growth of published articles per year	Cumulative percentage of no. of articles published per year	Annual growth rate (AGR) of published articles per year (%)
2011	2	1.7%	2	1.7%	–
2012	0	0.0%	2	1.7%	–
2013	1	0.8%	3	2.5%	–
2014	3	2.5%	6	5.1%	200%
2015	2	1.7%	8	6.8%	−33%
2016	2	1.7%	10	8.5%	0%
2017	3	2.5%	13	11.0%	50%
2018	6	5.1%	19	16.1%	100%
2019	6	5.1%	25	21.2%	0%
2020	2	1.7%	27	22.9%	−67%
2021	7	5.9%	34	28.8%	250%
2022	15	12.7%	49	41.5%	114%
2023	18	15.3%	67	56.8%	20%
2024	36	30.5%	103	87.3%	100%
2025	15	12.7%	118	100.0%	–
Total	118	100%	–	–	–

and Lu (2015) also expanded the research to surfing, exploring the link between recreational participation, flow experience, and overall well-being. They really emphasized the role of flow as a key mediating factor. The second most prominent area during this time involved cultural tourism, events, and service interactions. A good example is a study by Lee (2019) on the gamification of cultural heritage sites. Another area that stood out was foundational work in general tourism contexts and early ideas about combining technology with the field, with notable contributions from Gao and Bai (2014) and Huang et al. (2013).

Boom & Popularization Phase (2021–2024): Starting in 2021, the number of publications suddenly and sharply increased. The annual growth rate (AGR) reached a record 250% in 2021, followed by impressive growth rates of 114% in 2022 and 100% in 2024. In just four years, 76 articles were published, accounting for 64.4% of all research works in the 15 years. The peak year was 2024 with 36 articles, a number higher than the total number of articles from the previous 10 years combined. The timing of this boom points directly to the COVID-19 pandemic as the primary catalyst. The forecast for 2025, when the authors collected data only up to June 2025, recorded 15 articles, still higher than any year during the latency period. This phase is associated with the explosion of technologies such as virtual reality (VR) and augmented reality (AR), Tourism Live Streaming (TLS), the metaverse, and gamification. This context is suitable for a global industry that is in an existential crisis and desperately needs alternatives to physical tourism. The disruption caused by COVID-19 has created an immediate market demand for travel experiences that can be consumed remotely.

4.2. Most relevant authors

Lotka's law, a bibliometric principle, describes the distribution of scientific productivity among authors, typically following an inverse square law, where a small number of authors contribute a large portion of the publications in a field (Smolinsky, 2017), and is visually represented in the bibliometrix tool (Aria & Cuccurullo, 2017). From Lotka's law distribution of productivity in Fig. 1, it can be seen that to understand the main directions, fundamental theoretical frameworks, and emerging trends of the field, it is essential to focus on an in-depth analysis of the core author group, who are the most productive. They set the tone for academic discussions, set the standard for research methods, and point the field in the right direction. This report focuses on the notable author group identified by the bibliometric tool, as illustrated in Table 5. A close look at their work not only clarifies what each person did but also helps create a comprehensive knowledge map of this area of research.

"Albayrak, Tahir", "Huang, Xiaoting", and "Liu, Chun", each have three articles in the dataset. Following them is a group of authors with two papers, forming an important second tier in the productivity hierarchy. However, productivity is only a part of the story. Table 5 provides a more multidimensional view, showing not only the quantity but also the timeline and impact of these contributions.

A deeper analysis of these data reveals a significant difference between recent productivity and long-term scientific impact. Although Albayrak, Huang, and Liu were the most productive in terms of the number of articles, authors such as "Gao, Lingling", "Huang, Yu-Chih", and "Kim, Myung" had significantly higher total citations (TC) and citations per year (TCpY) for their fewer articles in the dataset. Specifically, two papers by Kim, Myung from 2017 and 2019 accumulated 99 and 363 citations, respectively. "Gao, Lingling", and "Huang, Yu-Chih" also brought back 336 and 359 citations, respectively, for their two publications. This discrepancy is not a contradiction but rather an indicator of the developmental structure of a scientific field. This shows that productivity (number of articles) is a measure of output and active participation in current discussions.

In contrast, impact (number of citations) is a measure of influence and reception by the scientific community. The data shows that Bai and Dematos wrote seminal or highly influential articles that shaped discussions and provided a conceptual foundation for the field. At the same time, today's most productive writers are building on this foundation and seeking new avenues to explore. A thorough analysis must acknowledge both factions: those who establish the foundations and those who construct new edifices. The structure of this report will reflect this, with separate sections for both the most productive and the most influential authors, whose work forms the basis for the entire field.

Albayrak, Tahir is a renowned expert in destination management, tourism marketing, and understanding tourist behavior. He teaches at Akdeniz University in Turkey. His academic record demonstrates a significant and lasting impact, as evidenced by his high H-index (37 overall, 34 in the last 5 years). This proves his place in the larger tourism research community. In the field of flow experience research, Albayrak's main contribution is his thorough study of its causes. He identifies flow as an essential mediating variable linking environmental stimuli to favorable tourist outcomes. His seminal works have explained this connection in-depth, for instance, the work, "How to create flow experience during travel: The role of destination attributes". This study is foundational to his work, using a quantitative approach with 938 tourists.

The study conducted by Karasakal and Albayrak (2022) empirically established that a tourist's perception of a destination's attributes (such as amenities or natural scenery) substantially affects the elements of flow (focused attention, time distortion, and diminished self-awareness).

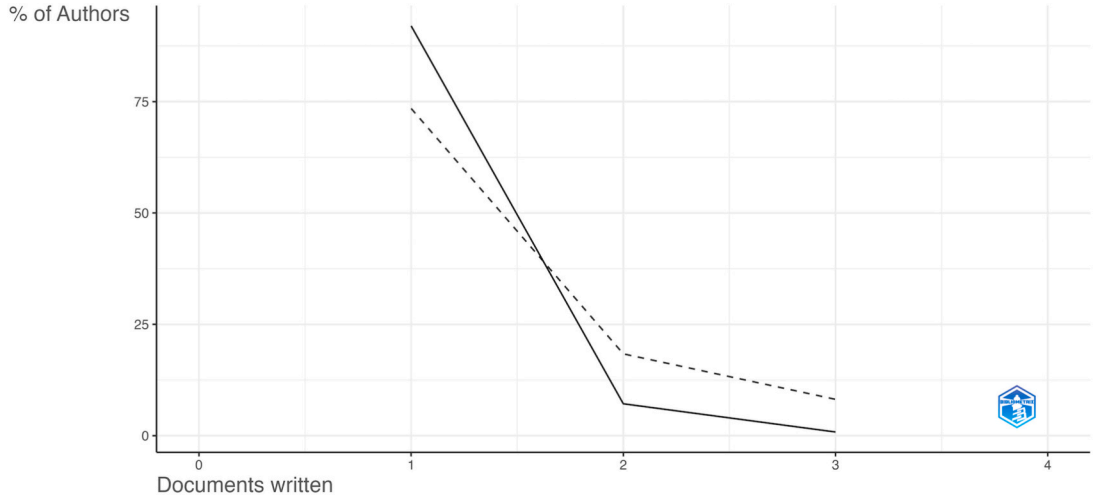


Fig. 1. Author productivity through Lotka's law. Source: visual map extracted from Bibliometrix.

Table 5
Profiles of leading authors in flow experience research in tourism. Source: author's synthesis from analysis results on Bibliometrix.

Author name	Total documents	Publication period	TC	TCpY	Main research themes/keywords
Albayrak, Tahir	3	2021–2024	77	19.45	Destination attributes, mobile applications, customer behavior, scuba diving, service quality
Huang, Xiaoting	3	2020–2023	95	22.53	Virtual reality (VR), psychophysiology, children's tourism, arousal, time perception
Liu, Chun	3	2021–2024	56	17.03	VR, psychophysiology, livestreaming, e-commerce, self-service technology (SST)
Gao, Lingling	2	2014–2017	336	30.97	Online travel agencies (OTA), virtual communities, online behavior, website atmospherics
Frash, Robert	2	2018–2019	47	6.34	Serious leisure, motorcycle tourism, travel intention
Caber, Meltem	2	2021–2024	56	14.20	Scuba diving, mobile applications, customer behavior, storytelling (close collaboration with Albayrak)
Chang, Hsuan	2	2022–2022	78	19.50	VR marketing, destination image, telepresence, youtube
Cheng, Tien-ming	2	2015–2024	60	5.86	Serious leisure, surfing, recreational involvement, safety climate, well-being
Dematos, Nelson	2	2021–2024	76	17.90	Conceptual review, leisure experience, novelty, memories
Huang, Yu-chih	2	2013–2023	359	40.95	Virtual reality (VR), 3D virtual world, technology acceptance, surfing
Karasakal, Sezer	2	2022–2024	31	10.25	Mobile apps, destination attributes, customer citizenship behavior
Kim, Myung	2	2017–2019	461	62.71	VR tourism, hedonic motivation, mobile social networks

TC: Total Citations.
TCpY: TC per Year.

These factors may make tourists happy. This finding provides a clear management direction, indicating that destination managers can enhance tourist satisfaction by improving destination attributes to facilitate the flow of tourists. Albayrak's more recent work uses his model on the digital-physical interface. It states that the quality of a mobile app (the stimulus) can influence the number of customers who come in and their level of engagement (the organism), which can lead to customer citizenship behavior (the response). This demonstrates a direct connection between flow experience and the long-term benefits tourism businesses can derive from it. This demonstrates that a well-designed app is not only a tool for accomplishing tasks but also a means of planning experiences (Caber et al., 2024).

One of the most interesting aspects of Albayrak's work is the frequent collaboration he shares with Caber, Meltem and Karasakal, Sezer. This is evident in several publications they have collaborated on. This partnership is a strong and lasting force in tourism research in Turkey, as it collaborates to address complex issues in consumer behavior. In general, the studies in this section, especially those by Albayrak, have consistently shown that flow is not the ultimate goal but rather an important psychological mechanism in the process. It is like the engine that turns external things (such as good service or beautiful scenery) into beneficial outcomes (such as satisfaction, loyalty, positive word of mouth, and happiness). Traditional service models might be a straight line from service quality to satisfaction. Nevertheless, Albayrak's work introduced the concept of "flow experience" as an essential intermediate step (destination attributes → flow experience → satisfaction).

This entirely reframed the objectives of tourism managers. Good service was no longer the only goal; designing experiences that could genuinely induce a state of flow was also a goal. Value was genuinely co-created in the visitors' minds while they were in this state of flux. This has important practical implications for destination managers and tour operators, who should focus on minimizing distractions, balancing challenges and skills, and creating the right immersive environments.

This group comprises researchers who have effectively applied flow theory in real-world tourism settings, including destinations, services, and physical activities. Tahir Albayrak, his frequent collaborator Caber, Meltem and Karasakal, Sezer have organized the role of destination attributes and service quality in creating flow. Another group of authors, including Frash, Robert and Cheng, Tien-Ming, has looked into flow in the context of "serious entertainment". Their studies Frash Jr et al. (2018), Frash Jr and Blose (2019), Cheng and Lu (2015), and Cheng et al. (2024) show that deep involvement and passion are strong signs of good experiences.

Gao, Lingling, was among the earliest contributors to the dataset (2014–2017) and had the highest citation impact by a significant margin (TC = 336, TCpY = 30.97). His work was foundational in applying flow theory to the development of the first wave of digital travel platforms. His significant works laid the foundation for understanding user experience in online travel, such as the highly cited study by Gao and Bai (2014) (229 citations), which established an important link between website design related to atmospherics and flow creation. Bai and his research team have demonstrated that factors such as visual design, usability, and interactivity of a website can induce a state of flow in users, and this state subsequently influences their online consumer behavior (booking intention). Next, in his 2017 work, he applied flow theory to explain sustained user engagement and participation in online communities. This indicates that when community members experience flow while interacting within the community (through reading and sharing travel stories), they are more likely to continue participating (Gao et al., 2017). This was an important precursor to modern social media studies. It can be seen that Bai and his colleagues' research was ahead of its time, recognizing that the digital experience is not just about functionality but also about emotion and immersion. The authors provide an experimental basis demonstrating that flow theory is a powerful analytical tool for deciphering user experience in the context of nascent online travel.

Methodological pioneers include the collaborative duo of "Huang, Xiaoting" and "Liu, Chun", who formed a separate group driving the field into an entirely new objective paradigm. By pioneering the use of psychophysiological methods in a VR context, they transformed the study of subjective states into a measurable science, opening up new possibilities for rigor and depth in analysis. In the study by Huang et al. (2021), physiological data (respiratory rate) were used to measure arousal in children experiencing VR. The findings show that longer videos lead to lower levels of arousal and less enjoyment, providing objective, data-driven insights for designing VR content for specific demographic groups. This represents a significant step forward from simply asking children whether they enjoyed the experience. Another joint publication by the two authors is titled "Does the selection of

virtual reality video matter? A laboratory experimental study of the influences of arousal". This study continues to refine its model using both self-reported stimulation and physiological data (breathing rate and heart rate) to demonstrate how different VR video stimuli generate varying levels of flow (enjoyment, time distortion, and attention focus) and satisfaction. Liu and Huang (2023) directly addressed the practical question of content selection for VR marketing, providing objective evidence to guide decisions.

The move toward psychophysiological methods, as shown by the work of "Huang, Xiaoting" and "Liu, Chun", can be seen as a turning point in the study of travel experiences. This area is moving away from asking guests, "How do you feel?" (from what you said) to objectively measuring "How did your body react?" (physiological response). Traditional flow research methods rely on post-experience questionnaires, which are susceptible to memory decay, social desirability bias, and difficulty in expressing complex emotional states. Conversely, psychophysiological measurements (such as heart rate, skin conductance, electroencephalography, eye tracking) record autonomic, unconscious responses in real-time.

The next group of scholars is the new media explorers, who have applied flow theory to the latest technological frontiers. Liu, Chun (in his independent work) has started to investigate flow in the context of live-streaming and TikTok, adapting this theory to fast-paced, commercial media environments. Chang, Hsuan, on the other hand, uses flow as a lens to evaluate the effectiveness of VR as a marketing tool, directly linking the psychological experience to business results.

Liu's groundbreaking work in this area is "Livestreamed tourism shopping: Consumers' attitudes and behaviors", which employs a sequential mixed-methods approach to develop a comprehensive model. In this model, the appeal of the source (including the streamer, the product, and the platform) and information quality act as the "Stimulus". The flow state (including elements like control, enjoyment, and curiosity) and cognition act as important intermediary factors of the "Organism". These factors then convert the stimuli into consumer attitudes and purchase intentions (Response) (Liu et al., 2024). This work is one of the first to systematically and multi-dimensionally apply flow theory to this specific and rapidly growing commercial context. It extends the concept of flow beyond purely recreational activities to include commercial settings, where flow is not just a desired experience but a powerful tool to drive purchasing behavior. Liu's research suggests that an engaging, immersive, and reliable live-streaming experience has the potential to place viewers in a state of flow, which reduces their risk perception and reinforces their transaction intention. This provides practical functional implications for tourism operators who wish to leverage this new sales channel.

Chang and Chiang (2022) employed flow as a theoretical framework to assess the efficacy of virtual reality (VR) as a destination marketing instrument, concentrating on the impact of immersive experiences on the perceptions and attitudes of prospective visitors. This study looked into this connection in great detail. It found that components of the flow experience in VR, particularly friendliness and telepresence, had a positive impact on destination image. This improved destination image, in turn, had a significant indirect impact on changing users' attitudes toward the destination.

4.3. Most relevant sources

In this section, the study identifies nuclear journals according to Bradford's law, a bibliometric principle that allows for the determination of core journals in a specific research field. According to this law, a small number of journals account for a large proportion of the most important articles. Applying this principle to the dataset on flow experience in tourism, the analysis clearly identifies a group of six "Zone 1" journals that are considered the intellectual core of the field.

This core group of journals includes the Asia Pacific Journal of Tourism Research, Journal of Vacation Marketing, Sustainability (Switzerland), Tourism Management, Tourism Review, and Current Issues in Tourism. As shown in Table 6, this small group, although representing a fraction of the 56 publishing journals, accounts for a large volume of articles (40 out of 118), comprising more than one-third of the total scientific output.

The existence of such a clear core group suggests that this field is not fragmented; instead, it has a distinct intellectual center, which helps new researchers quickly identify the most important works. Simultaneously, with over 30 journals contributing just a single article (for example, Foods, Journal of Global Fashion Marketing, and Event Management), this demonstrates the flexibility and broad applicability of the flow concept. It has been tested and applied in various specialized tourism contexts and related fields, such as technology and psychology, demonstrating the concept's robustness and interdisciplinary appeal.

Fig. 2 provides a closer look at the 10 most influential journals based on their publication output across all 56 journals in the research field. The Asia Pacific Journal of Tourism Research and the Journal of Vacation Marketing led the way, each with eight articles. Following closely are Sustainability (Switzerland) with seven articles, and Tourism Management and Tourism Review, each with six articles. Current Issues in Tourism (five articles) and the Journal of Travel and Tourism Marketing (five articles) are both prestigious and highly influential journals in the tourism industry. The strong presence of articles on flow experience in these journals confirms that this is an important and influential research topic in tourism. However, this study also utilizes the local H-index to provide a more accurate representation of the quality and impact of the works. This index examines both the productivity and frequency of the article citations in a specific dataset. The analysis shows that the Asia Pacific Journal of Tourism Research, Journal of Vacation Marketing, and Tourism Management are tied for first place with a local H-index of 6. This means that every one of these journals has at least six articles, and each of those articles has been cited at least six times in the same dataset. This result is important because it shows that the articles published in these three journals are not only numerous but also of high quality, often cited by scientists, and have a significant effect on academic discussions.

The journal Tourism Management is a theoretical and methodological pillar, as it boasts a high Local H-index, ranking it among the top three journals in its field. Tourism Management has established itself as a leading forum for disseminating research that advances the knowledge base of the field. Citation numbers do not just demonstrate this role, but also the very nature of the papers published.

A review of the top 10 most-cited articles shows that two foundational works were published in this journal: the study by Wu and Liang (2011) on adventure tourism and the one by Huang et al. (2013) on 3D virtual worlds. These articles share a common trait: they established early theoretical models, systematically connected flow theory to tourism contexts, and tested them using rigorous quantitative methods. They did not just apply the concept of flow; they also explored its antecedents (skills and challenges) and consequences (satisfaction and intention to adopt technology). Meanwhile, the Asia Pacific Journal of

Table 6
Overview of core journals in flow experience research in tourism. Source: author's synthesis from analysis results on Bibliometrix.

Journal name	Total articles	Local H-Index	Bradford's Law Zone
Asia Pacific Journal of Tourism Research	8	6	Zone 1
Journal of Vacation Marketing	8	6	Zone 1
Sustainability (Switzerland)	7	4	Zone 1
Tourism Management	6	6	Zone 1
Tourism Review	6	4	Zone 1
Current Issues in Tourism	5	5	Zone 1
Journal of Travel and Tourism Marketing	5	4	Zone 2

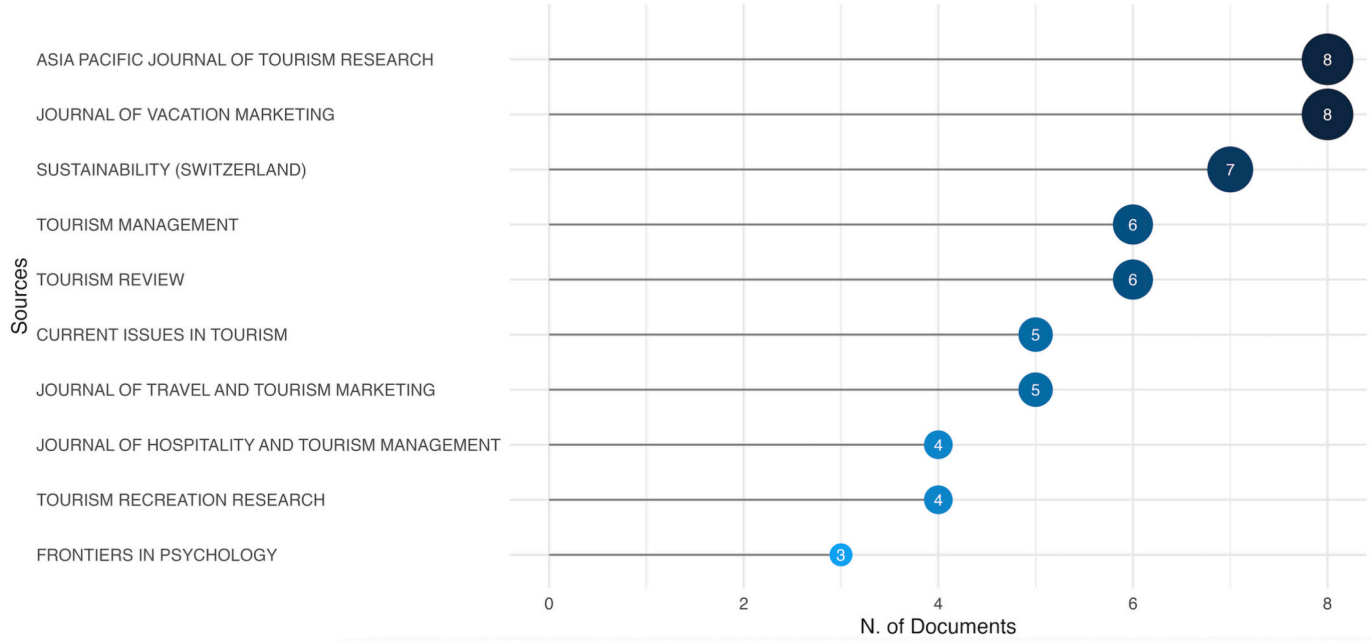


Fig. 2. Most relevant sources. Source: visual map extracted from Bibliometrix.

Tourism Research and the Journal of Vacation Marketing are driving forces behind applied and contemporary research. Similar to Tourism Management, these two journals have the highest influence in the field, as measured by their impact indices. They also best reflect current research trends and popular topics. The explosive growth in publications from these two journals after 2020 suggests that they are the primary forums for research on technology applications in tourism, a trend likely accelerated by the pandemic.

4.4. Most-cited documents

Analyzing document citation indices is an effective method for identifying foundational and highly influential works that shape the knowledge structure of a research field. Table 7 below presents the 10 most cited publications globally on the topic of flow experience in the tourism industry, providing a quantitative overview of the field's intellectual landscape.

With 362 citations and the highest citations per year (TC per Year) rate on the list (51.71), the work by Kim and Hall (2019) has a significant impact on the most recent research direction regarding the intersection of technology and tourism. This study proposes and tests a Hedonic Motivation System Adoption Model (HMSAM) that integrates core factors, including perceived enjoyment, flow state, and subjective well-being, to explain the intention to continue using virtual reality (VR) tourism. The central finding of this study is the decisive mediating role of flow state, which acts as a psychological mechanism that transforms initial interest into deeper outcomes, such as subjective well-being and long-term intention to engage with technology. A particularly insightful finding was the regulatory role of real-world experience: the relationship between the usefulness of the VR platform and the flow state experienced by users differed significantly between those who had visited the destination in reality and those who had not. This indicates that virtual experiences are not isolated but are influenced by users' prior memories and knowledge.

Analyzing these core publications reveals a clear trend: the dominance of technology. Flow theory, which was developed to describe deep engagement with an activity, has found a perfect laboratory in digital environments. In these environments, variables such as interaction, liveliness, and control can be measured and manipulated much more easily than in real-world travel experiences, which are complex and unpredictable. Therefore, flow research in tourism has been driven and shaped by the evolution of consumer technology, from 3D virtual worlds, websites, and mobile applications to VR.

Theoretically, this study upgraded traditional technology acceptance models (such as TAM) by shifting the focus from functional (utilitarian) aspects to hedonic aspects, which are more suitable for recreational products such as VR tourism. Notably, this is one of the first studies to include subjective well-being as an important outcome variable, opening up a new direction for evaluating the impact of virtual tourism on not only behavior but also consumers' mental health. Practically speaking, this study sends a clear message to marketers and destination developers: for a VR travel application to be successful, it must be designed to maximize enjoyment and immersion, thereby creating a state of flow, enhancing well-being, and encouraging users to return to the platform, ultimately boosting their intention to visit the destination in the real world.

Huang et al. (2013) laid the groundwork for understanding hedonic experiences in virtual tourism environments. This is a foundational work with 307 citations, and despite being published in 2013, it continues to average 23.62 citations per year. As mentioned earlier, this paper was published in Tourism Management, a journal with a high impact on flow experience in tourism. This is also reflected in the publication's content, as it was one of the first to highlight the limitations of the traditional Technology Acceptance Model (TAM) when applied to the entertainment industry. The authors argue that the core functional elements of TAM, such as “perceived usefulness” and “perceived ease of use”, were not sufficient to explain user behavior in an environment where the main goals are entertainment and experience, not work performance.

The most significant theoretical contribution of this study was the extension of TAM by integrating a range of hedonic constructs, including enjoyment, emotional involvement, positive emotions, and especially, the flow experience. This study clarified the role of emotional and psychological factors in technology acceptance models in the tourism field. This paved the way for a wave of later studies, including the work by Kim and Hall (2019), that explored the complex facets of the interaction between technology and tourists more deeply.

Following Huang, Gao and Bai (2014) applied the SOR (Stimulus-Organism-Response) theoretical framework to the context of tourism websites. This study had a total of 229 citations, ranking third and averaging over 19 citations per year. The important contribution of this

Table 7
The 10 most impactful studies on flow experience in tourism, based on citation index. Source: author's synthesis from analysis results on Bibliometrix.

Rank	Reference	Total citations	TC per year	Core topic	Main theoretical contribution
1	Kim and Hall (2019)	362	51.71	Virtual Reality (VR) Tourism, Hedonic Motivation	Developed the Hedonic Motivation System Adoption Model (HMSAM), integrating flow and subjective well-being.
2	Huang et al. (2013)	307	23.62	3D Virtual Worlds, Technology Acceptance	Extended the Technology Acceptance Model (TAM) with hedonic constructs.
3	Gao and Bai (2014)	229	19.08	Tourism Websites, Online Consumer Behavior	Applied the S-O-R framework, positioning flow as the core psychological mediating variable.
4	Tan and Ooi (2018)	222	27.75	Mobile Tourism Shopping, UTAUT	Integrated Flow Theory into the UTAUT model and tested the role of demographics.
5	Kim and Thapa (2018)	216	27.00	Nature-based Tourism, Perceived Value, Sustainable Behavior	Connected perceived value with flow and extended outcomes to environmentally responsible behavior.
6	Chang (2014)	174	14.50	Tour Guides, Trust, Shopping Behavior	Examined the role of flow in the relationship between tour guide performance and tourist shopping behavior.
7	Wu and Liang (2011)	161	10.73	Adventure Tourism (Rafting), Satisfaction	Developed a model of flow experience formation in an adventure tourism context.
8	Gao et al. (2017)	107	11.89	Virtual Travel Communities, Sustained Participation	Integrated the IS Success Model and Flow Theory to explain long-term participation.
9	Kim et al. (2017)	99	11.00	Mobile Social Networks, Purchase Intention, Elderly	Studied the tourism-related purchase intentions of the elderly on mobile social networks.
10	Wu and Lai (2022)	89	22.25	Short Videos, TikTok, Behavioral Intention	Applied the S-O-R model to test the impact of short video factors on flow and travel intention.

study is that it clearly positions flow experience as an organism – the core psychological mechanism mediating the transformation of website design elements (Stimulus) into purchase intention and satisfaction (Response). Together, these two studies establish a foundation for flow research in pre-trip digital tourism experiences.

The two studies with over 200 citations were Tan and Ooi (2018) and Kim and Thapa (2018), which aimed to integrate or expand complex models in more diverse contexts. Tan and Ooi (2018) incorporated the concept of flow into the UTAUT (Unified Theory of Acceptance and Use of Technology) model within the context of mobile travel shopping, demonstrating the suitability of this theory on emerging technology platforms. Simultaneously, Kim and Thapa (2018) created an important bridge between the digital space and the real-world experience at the destination. This study showed that perceived value in nature tourism (a marketing factor) is an antecedent of inducing a flow experience (a psychological state), and this state not only causes loyalty (a business result) but also encourages environmentally friendly behavior (a sustainability consequence). These studies illustrate the maturity of the field, as flow is employed to describe a wide variety of phenomena, ranging from mobile commerce to ecotourism.

4.5. Thematic map analysis

Analysis of the bibliographic map of flow experience in tourism from 2011 to 2025 reveals a complex intellectual structure shaped by foundational themes and driven by dynamic research areas. By analyzing a Thematic Map divided into four quadrants based on centrality (horizontal axis) and density (vertical axis), we can identify core concepts, topics driving the field's development, specialized areas, and emerging themes. This section deconstructs the intellectual foundation of the research field by analyzing the topics located in the “Basic Themes” and “Motor Themes” quadrants. This will help us establish foundational concepts and identify the central topics that are shaping the field's developmental trajectory.

4.5.1. The foundational pillars: basic themes

Basic themes (bottom right corner) represent the fundamental concepts and theories of the field. They are highly central, meaning they have widespread connections and serve as a bridge for many different research areas. However, their low density suggests they are not subjects studied independently and in-depth, but rather serve as theoretical lenses or general contexts for exploring other issues.

The core of the entire research field is the theme cluster “flow experience” (located in the red cluster under basic themes). The data in Table 8 show that the highest centrality (Callon's Centrality) is observed among all clusters (0.894), but the development level (Callon's Density) is relatively lower (41.118, ranked 3rd out of 8). This suggests that “flow experience” is a central concept connected to many different topics; however, the studies in this cluster may not yet form a tightly linked and densely developed body of work, unlike the motivational themes. In the red cluster, “satisfaction” also emerged as an integral component, appearing 16 times. This close link is evident from the connection and shared membership in the same cluster in the co-occurrence network shown in Fig. 4. Satisfaction is often regarded as one of the most direct and important outcomes of achieving a flow state. Relevant studies often investigate flow as a precursor to tourist satisfaction, whether in adventure sports or virtual travel experiences. The presence of other important keywords, such as “virtual reality”, “destination loyalty”, “revisit intention”, “perceived value”, along with theoretical frameworks such as “SOR model” and “TAM” (technology acceptance model), further reinforces that this cluster represents the application of flow theory to predict important behavioral and marketing outcomes. This structure reveals a dominant research paradigm in which researchers focus not only on discussing flow theory in the abstract but also on applying it in practice. Instead, they operationalize it in technology-mediated environments (primarily VR) to measure its impact on established outcomes.

The prominent position of “virtual reality” (VR) within the core theme cluster “flow experience” makes it the second most central term in this cluster. This shows that VR serves as a primary, almost indispensable environment for contemporary studies on flow experiences in tourism. VR technology, which can create highly immersive and interactive environments, is a great way to study flow conditions such as

Table 8
Analysis of thematic clusters in flow experience in tourism research. Source: author's synthesis from analysis results on Bibliometrix.

Cluster	Quadrant	Relevance (Centrality)	Development (Density)	Cluster Frequency	Key Terms in Cluster	Role in the Research Field
flow experience	Basic Themes	0.894	41.118	154	satisfaction, virtual reality, perceived value, sor model	The conceptual nucleus of the entire field, closely linked with VR technology and core psychological outcomes like satisfaction.
tourism	Motor Themes	0.573	61.181	28	Tourism live streaming, Asia Pacific, tourism marketing	An overarching, well-developed, and highly connected theme, serving as a primary driver of research. The live streaming subfield is a strongly developing branch.
authenticity	Motor Themes	0.433	62.000	20	enjoyment, cognitive response, gamification, presence	A significant, highly developed motor theme, focusing on the psychological outcomes of the tourism experience and the role of authenticity.
travel intention	Basic Themes	0.279	47.449	23	sports tourism, destination, serious leisure, storytelling	A transversal theme connecting different areas, particularly sports tourism and serious leisure, focusing on behavioral intentions.
visit intention	Niche Themes	0.000	50.000	2	–	A very specialized and nascent theme with potential for development but currently isolated.
customer engagement	Niche Themes	0.000	62.500	4	influencer marketing	A specialized theme, highly developed internally but still isolated, focusing on influencer marketing.
cultural tourism	Emerging/ Declining Themes	0.167	33.333	3	–	A peripheral theme, potentially in its early stages of development or losing relevance in this specific context.
technology acceptance model	Emerging/ Declining Themes	0.067	20.000	5	–	A peripheral theme, likely being integrated into larger models rather than being studied independently.

presence, vividness, and telepresence. Research indicates that vividness and presence in VR are effective indicators of flow experience, which in turn leads to hedonic motivation and visual appeal, ultimately influencing the intention to visit.

The topic clusters of “sports tourism” and “travel intention” are located within the purple cluster in Fig. 4, clearly illustrating the classical flow model. In highly skilled and challenging activities such as “white-water rafting”, “scuba diving”, and “motorcycle tourism”, the flow state is studied as a result of the optimal balance between the participant's physical skills and environmental challenges. The concept of “serious leisure”, found in the “travel intention” cluster, is also integrated as an important predictor of flow in these contexts, indicating that participants' commitment and skill investment are prerequisites. Here, the primary trigger is the balance between one's skills and challenges. The problem is that the purple cluster is located almost at the center of the thematic map. This position is biased toward basic themes, but this is not obvious when looking at the data in Table 8. This cluster appears to connect specialized tourism fields, such as “sports tourism” and “serious leisure”, with broader behavioral outcomes. Additionally, it serves as a bridge between stimuli (such as storytelling)) and tourists' final responses.

4.5.2. *Drivers of progress: motor themes*

Motor themes, located in the top-right quadrant of the thematic map, are defined by both a high level of development and strong relevance. These are the well-established, tightly connected research areas that act as the main drivers for the entire field. The thematic map shows they focus on core psychological aspects of the travel experience and the technological and market platforms shaping modern tourism.

The orange “authenticity” cluster, with a high density (62.000) and strong centrality (0.433), is a good illustration of a motor theme. This shows that it is a critical, well-developed, and cohesive research area that is driving the field forward. The two main elements of this cluster, “enjoyment” and “cognitive response”, indicate that the study being conducted here extends beyond authenticity. It delves into its psychological mechanisms - how a perception of authenticity leads to the emotional outcome of enjoyment and activates a tourist's cognitive processes. The direct link between perceived authenticity, enjoyment, and the subsequent emergence of a flow state is a clearly established pathway in academic literature, often referred to as the “perception – enjoyment – flow” pathway. This is the central driving force of this topic.

Interestingly, “tourism live streaming” (the light blue cluster) has emerged as a powerful motor theme, as seen in Fig. 3, with high density (61.181) and strong centrality (0.573). Its position in the top-right quadrant and its inclusion in the “tourism” cluster indicate its rapid and recent development into a central research topic. Live-streaming platforms have recently been shown in studies conducted by Ye et al. (2022), Zheng and Fu (2023), and Liang et al. (2024) to serve as a core instrument for engaging and authentic, flow-inducing experience shaping travel intentions of the viewers.

Although the keyword “tourism” might seem generic, the most important and central terms in this cluster are concrete: “tourism live streaming” and “Asia Pacific.” This finding provides a powerful insight that general topics on “tourism” within the flow experience context are now being driven almost entirely by research on live-streaming, with a clear geographic focus on the Asia-Pacific region. The flow experience is the key psychological state that connects these two pillars. Live streaming is effective precisely because its features (interactivity, immediacy) can create a flow state, which then satisfies a tourist's underlying desire for an engaging and authentic-feeling experience. So, flow is not just an outcome; it is the mediating mechanism that makes tourism technologies successful.

4.5.3. *Specialized fields: niche themes*

Niche themes are located in the upper-left quadrant and represent highly specialized research areas. They are characterized by a very high density, indicating a strong internal structure and extensive development, but very low centrality, indicating relative isolation from the core topics of the field.

The “customer engagement” cluster (brown) is an example of a niche theme. This cluster has the highest density of all clusters (62.500) but has the lowest possible centrality (0.000). This suggests that it is a well-developed, in-depth, and internally coherent research area. However, it is largely isolated from the central theoretical discussions in the rest of the field. The keyword “influencer marketing” is a key component of this cluster, indicating that researching “customer engagement” in this context is almost synonymous with studying influencer marketing. Studies are increasingly exploring how an influencer's characteristics (such as aesthetics or parasocial interaction) can create a flow experience for followers, which in turn drives purchase intentions. This connection refers to the maturation of this research area, in the sense that the discipline is transcending the mere examination of advertising impact (reach and clicks) and approaching an understanding of the underlying psychological mechanisms (eliciting an immersive flow

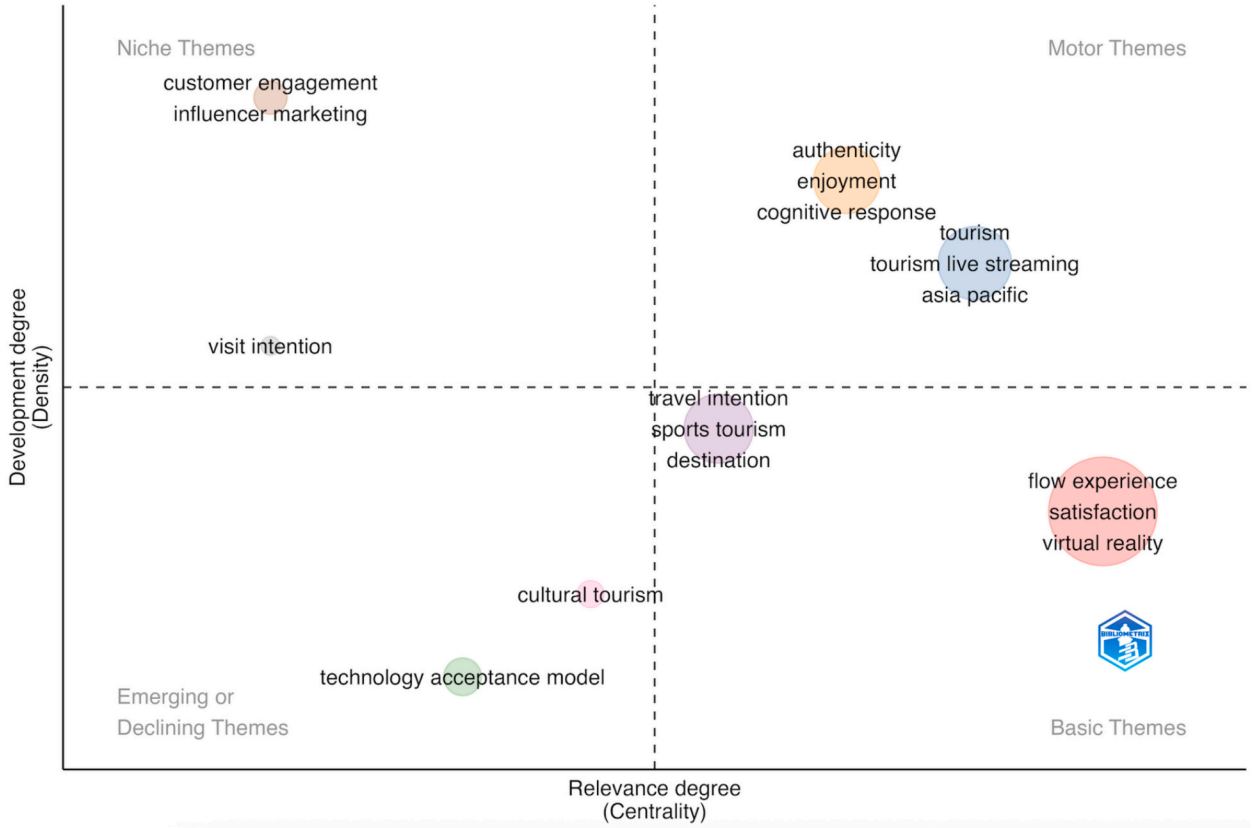


Fig. 3. Thematic map based upon author's keywords (2011–2025). Source: visual map extracted from Bibliometrix.

state) that drive influencers to be impactful. This change in emphasis may render this topic more core in the future, as it speaks more fundamentally to basic psychological theory.

"Visit intention" is another specialized topic that has a lot of development (density: 50.000) but no centrality (0.000), just like "customer engagement". This makes it a clear, but separate, area of research. This small group of themes reveals a significant gap between marketing applications and theoretical foundations. There is a wealth of research on topics such as "customer engagement" and "influencer marketing", which have well-established internal structures. However, their low centrality suggests that they do not effectively tap into or contribute to fundamental "flow theories" or the psychological drivers behind the "authenticity" motor theme.

4.5.4. Emerging or declining themes

The subjects in the lower left corner of the map are about new or changing areas. They have both low density and centrality, which means they are not well-developed and are rarely discussed in mainstream conversations. However, their presence could indicate that the field's knowledge structure is evolving and moving in new directions. The "technology acceptance model" (TAM) topic cluster falls within the quadrant of emerging or declining topics. The data shows it has both low relevance (0.067) and low development (20.000), indicating that it is not a primary research driver in this field. However, this does not mean that TAM is declining in importance. Instead, it is being integrated as a foundational component into more complex models to explain the adoption of new travel technologies.

This trend reflects a significant theoretical evolution in the field. Initial studies on technology adoption often focused on utilitarian aspects: is the technology helpful (perceived usefulness) and easy to use (perceived ease of use)? This is the core of TAM. However, tourism is fundamentally a hedonic and experiential activity. Directory data shows that researchers are increasingly recognizing that for travel technologies, adoption depends not only on usefulness but also on the quality of the experience it provides. Flow theory provides a theoretical framework and language for measuring the quality of that experience. Therefore, this field is shifting from a functional acceptance model to a holistic experience model. Many recent studies have explicitly integrated TAM with Flow Theory better to understand behavioral intentions in VR and Metaverse environments. In these integrated models, TAM variables are often positioned as either antecedents or outcomes of flow state, creating a more comprehensive model of user experience and acceptance.

"Cultural tourism" also appears in a similar quadrant but leans toward more emerging themes (density: 33.333, centrality: 0.167). This means that while flow has been extensively studied in adventure tourism and adventure sport, its application in the more reflective and educational context of cultural and heritage tourism is a relatively new and less established discipline. Here, flow is not built up through bodily challenges but through cognitive and emotional absorption within artistic, historical, and cultural stories. The precipitating factor in this instance is cognitive involvement and presence within the narrative. Technology, and more specifically VR and AR, plays a pivotal role in enhancing visitor numbers at sites of cultural heritage. The two technologies can bring about a historical context, offer interactive stories, and create a greater presence, thereby creating a state of flow. Its application to the preservation and management of these sites is also being reconsidered here, where not only are they conserving, but they are also generating virtual experiences that can induce a state of flow.

4.6. Frequency of words over time

In this section, the study conducts a long-term word frequency analysis to examine the development and impact of key terms over time in a particular research area. This analysis will enable us to identify which words appeared most frequently in academic papers during

different periods, providing deeper insight into the trends and changes in the terminology used by the research community.

Fig. 5 shows that the bibliographic data show a slow but steady increase in core keywords such as "flow theory," "satisfaction," and "enjoyment". This period is not characterized by a quantitative explosion; rather, it is a process of building a solid intellectual foundation, in which the academic community has attempted to validate and integrate the basic principles of flow theory into the specific context of tourism. The keywords "flow" and "flow state" appear early and have a steady, albeit low-key, presence in the graph, but they also show a dramatic spike in 2022. This underscores its role as a key ingredient and an absolute necessity for achieving a flow state. More importantly, the appearance of the keyword "flow experience" as a separate term on the chart and in the data indicates the shift in focus from "flow" to user experience. This keyword experienced a significant breakthrough in 2021 and continued to increase exponentially in the following period. The keyword "destination loyalty" also appeared early, although at a lower frequency. This reflects a logical chain established in studies of flow experiences, which leads to satisfaction, which is the primary driver of loyalty. The most striking feature of the directory graph is the exponential growth of the keyword "virtual reality", which began around 2018–2019 and accelerated strongly thereafter. This explosion does not simply represent the emergence of a new topic. However, it represents a fundamental shift in methodology and research objectives from observing and measuring flows in natural environments to actively designing, engineering, and creating flows in technologically mediated environments. The graph analysis shows that "authenticity" is an emerging but rapidly accelerating topic, reflected in its late but steep rise. This growth is viewed as a response to and amplification of the experience economy, marking a shift in tourist demand from simple hedonic or immersion-type experiences to those that are more authentic, meaningful, and deeply transformative.

4.7. Keywords occurrence

As explained earlier, the current study applies co-occurrence analysis with the VOSviewer tool to enable identification of the relationships between keywords within science reports in order to reveal research trends and thematic clusters (Li & Wei, 2022; Sedighi, 2015). In the realm of tourism, this methodology can delineate the research framework of flow experiences through the examination of the frequency and interconnections of keywords about flow, tourism activities, and psychological effects. The analysis results suggest that there are four main topic clusters, which are clearly shown by the colors in Fig. 5.

4.7.1. Cluster 1 – green: the core of experience - the psychology of tourist engagement

The green cluster is the theoretical center of the map, focusing on the psychology of the experience of the visitor. This cluster examines what immersion is, its cause, and how new technologies impact it. This group examines what immersion is, its causes, and how new technologies are impacting it. The "flow experience" node, positioned in the middle of the green cluster, is the largest, indicating that it is particularly crucial to this field. Not only the most common occurring keyword, but also the middle idea upon which the psychological study of holiday experience is based. It links basic psychological principles like "flow state", "enjoyment", "psychology", and "virtual reality". VR is not just another way to travel; it is a technology designed to create immersive, engaging environments that are ideal for achieving a state of flow. One of the strongest and most important linkages in this group is between "flow experience" and "authenticity", on the other side. The map suggests that seeking out real experiences is not only a goal in itself, but also an important means to achieve a state of flow.

4.7.2. Cluster 2 – red: strategic applications - from tourist behavior to destination management

While the green cluster illustrates the "why" and "what" of the flow experience, the red cluster deals with the "how" and "what for". This cluster is grounded in managerial concepts and applications, illustrating how psychological theory is applied in practice to achieve strategic goals in the tourism industry. The red cluster shows that "flow theory" is not an isolated concept but a theoretical lens that is widely applied to the fields of "marketing", "tourism development", and "tourist destination management". The close connection between these terms shows that researchers and practitioners are becoming more aware that designing experiences that generate flow on purpose can give them an edge over their competitors. This could involve running heritage tourism sites to engage more people, rather than just letting them passively enjoy them, or creating adventure tourism products that are carefully designed to cater to different skill levels, which would help increase participation. The close association between flow theory and the Technology Acceptance Model (TAM), proposed by Davis (1989), is a foundational model for explaining technology acceptance, based primarily on two core beliefs: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). It is more than just a combination of two theories; it represents the recognition that, in tourism, the quality of the experience is a core component of utility.

4.7.3. Group 3 – gold: the link between "flow" and "satisfaction" - a core principle of the travel experience

The proximity of the nodes and the thickness of the connecting lines between them on the VOSviewer map indicate a very high co-occurrence frequency and a direct and strong relationship. This visual shows that the link between "flow" and "satisfaction" is one of the most robust and frequently validated findings in this field of research.

4.7.4. Cluster 4 - blue: emerging lines and contextual shades

The blue cluster, including "China", "perception", and "tourism management", is a notable finding. The formation of this distinct, geographically concentrated cluster indicates a significant volume of material specifically focused on the Chinese context.

4.8. Co-citation of authors

Fig. 7 presents a co-authorship analysis, highlighting the most important and relevant collaborations in this bibliographic research paper of ours. This analysis allows us to visualize the connections between authors and explore the collaborative networks that have emerged. The principle of this method is very intuitive: when two authors are cited together in a third research work, they form a co-citation link. The higher the frequency with which the two authors are cited together, the stronger their academic and conceptual connection is considered to be. This shows that the scientific community considers their works relevant, complementary, or belonging to the same school of thought. Author co-citation analysis provides a comprehensive, structured, and multi-dimensional overview of the research field of flow experience in tourism. It not only identifies the foundational authors and theories but also sheds light on the methodological evolution and the differentiation into various research disciplines.

4.8.1. Cluster 1 (red) - foundational pillars: original theory and classic methodology

Cluster 1, shown in red, is the largest and most central cluster on the intellectual map, which shows its fundamental role in the entire field. The center of this cluster is the node "Csikszentmihalyi M.", whose size is superior to that of the other authors, confirming his position as the father of Flow Theory, the core theory around which all research in the field revolves.

However, this cluster is not limited to the Flow Theory. The presence of other influential authors indicates that a rich "theoretical ecosystem"

was formed from the beginning. “Pine B.”, co-author of the classic “The Experience Economy”, provides the macroeconomic context, arguing that businesses, including the tourism industry, must move from selling services to staging memorable experiences to create value. Methodologically, the work of “Fornell C.”, and “Anderson J.”, is vital. The classic work by Fornell and Larcker (1981) on the evaluation of structural equation models (SEM) laid the foundation for the application of rigorous statistical methods in this field.

4.8.2. Cluster 2 (blue) - focus on experience: operationalizing and measuring the tourist experience

Cluster 2, represented in blue, serves as an intermediary between the theoretical ideas of Cluster 1 and their practical implementation in the travel industry. The researchers in this cluster are dedicated to “operationalizing” rich experience concepts, making them variables that can be measured, evaluated, and controlled. It is here that theory comes closest to the empirical data. Another notable research direction in this cluster is the exploration of technology-mediated experience. “Backman S.”, and especially “Tussyadiah I.”, are pioneers in researching the virtual reality (VR) tourism experience. Their work not only applies Flow Theory to understand user immersion but also integrates concepts such as presence (understood as the feeling of actually being there) to explain how VR can change tourists' attitudes and intentions toward visiting a real-world destination.

One of the central figures in this cluster is “Oh H.”, who has influential research on measuring “experience economy” concepts. Focusing on the bed-and-breakfast lodging sector, Oh et al. (2007) conducted preliminary qualitative research and a field survey to gather data from B&B owners and guests, thereby developing and testing their proposed model of experience economy concepts. Similarly, “Chen C.”, made important contributions to the study of the relationship between experience quality, perceived value, satisfaction, and behavioral intention, especially in the context of heritage tourism.

4.8.3. Cluster 3 (green) - approach modern quantification

This cluster is clearly defined by the influential trio of authors in the field of methodology: “Hair J.”, “Ringle C.”, and “Sarstedt M.”. The close connection of this methodological team of authors with leading experts in the application of technology in tourism, such as “Buhalis D.”, (expert in e-tourism and smart tourism) and “Wang Y.”, (researching new technologies and social media), is not coincidental. This reveals a clear trend: as travel becomes increasingly digitized, researchers are confronted with increasingly complex theoretical models that incorporate multiple latent variables, including vividness, interactivity, presence, flow experience, and behavioral intention.

4.8.4. Group 4 (yellow) - methodological pioneer school: advanced quantitative techniques

Cluster 4, which has the fewest authors, is critical because it shows how the field has “methodologically matured”. This cluster, which is coded in yellow and located in a relatively remote area, focuses on improving, refining, and making the quantitative methods used more rigorous. The central and defining author of this cluster is “Henseler J.” One of his most notable contributions is the proposal of the Heterotrait-Monotrait ratio of correlations (HTMT) criterion for assessing discriminant validity more rigorously than the traditional Fornell-Larcker criterion. His works are often highly technical and critical, pushing the research community toward higher standards of reliability and validity of results.

4.9. Trend topics

Fig. 8 shows the terms with trend topics. “Flow experience” with a frequency of 86 is the central and most influential topic in the entire field. The trend chart shows a growth and sustainability line, particularly one that experienced significant growth after 2020. This confirms that “flow experience” is not only a popular topic but also a field of research that is increasingly attracting interest and being explored in depth. Hot and emerging topics are pioneering research directions, attracting strong attention in recent times and promising to shape the future of the field, including “virtual reality”, “tourism”, “travel intention”, with frequencies of 16, 10, and 7, respectively. All three topics have shown a recent increase, reflecting the trend of focusing on the results and practical applications of flow experiences. This research not only describes the experience but also examines its impact on the tourism industry as a whole and the travel decisions of customers. Analysis of trending topics shows that the newest and most pioneering topic is “authenticity”. The frequency of appearance is still modest, but the research period is extremely new (focused on 2024 and expected to last until 2025). This shows that researchers are starting to explore the complex relationship between the flow experience (often created by technology) and the need for “authenticity” of tourists. This represents a new and potentially promising research direction.

5. Discussions

Based on the research results, including thematic and keyword co-occurrence analyses on flow experience in tourism, this study synthesized the main findings from the bibliometric analysis into a coherent knowledge structure, as illustrated in Fig. 9. The main content of this research framework was also thoroughly explored and analyzed to reflect the structure and dynamics of flow experience research in tourism. This framework separates the research field into three main pillars: (1) how we learn about the field; (2) what we know about the field; and (3) where we study the field. Additionally, in the discussion section, this study clarifies the research questions raised earlier in the overview.

5.1. RQ1: the trajectory of a field: from niche concept to mainstream phenomenon

The analysis of annual scientific output reveals a rather uneven developmental trajectory. It can be divided into two distinct periods, with the COVID-19 pandemic serving as a decisive turning point that propelled the field from niche interest to mainstream research.

The initial phase and slow growth (2011–2020): Annual scientific output data show a clear picture of how this research field has developed. During the first decade of the analysis, the number of published articles was relatively low and highly volatile. The annual growth rate (AGR) was unstable, with significantly negative growth years, such as −33% in 2014 and − 50% in 2016. This instability suggests that during this phase, flow experience remained a specialized theoretical concept, applied inconsistently and somewhat disjointedly across different tourism contexts, primarily in adventure and outdoor recreational activities.

The boom and mainstreaming phase (2021–2024): Starting in 2021, there was a considerable change. The number of publications skyrocketed, with the AGR increasing by 200% in 2021 and continuing to grow at an impressive rate in subsequent years. This rapid growth was not a natural process; the COVID-19 pandemic directly caused it, which was a global exogenous shock. The near-complete cessation of physical tourism worldwide has created an urgent and unprecedented need for new travel methods. In this situation, virtual reality (VR) tourism has rapidly evolved from a helpful marketing tool or novelty into a vital means for destinations to remain vibrant, resilient, and appealing.

As virtual tourism became a central focus, researchers and managers asked, “how can a virtual experience be interesting, real, and useful?”. Flow theory, which emphasizes complete immersion, focus, and enjoyment, is a valuable approach for understanding and enhancing the user experience in virtual environments. Therefore, the crisis in physical tourism directly fueled the boom in virtual tourism, which in turn created a wave of research applying flow theory to this new context.

The suggestions from Fig. 5 also indicate that the growth of the research area adheres to the growth stages outlined in Table 4. The keyword "Flow Experience" analysis using a trajectory model also shows that this keyword is most prominent and bare, with a steadily rising curve followed by a sharp acceleration. Fig. 10, which includes an analysis of the Web of Science (WoS) dataset used for comparison with the primary analytical dataset from Scopus, also yields similar conclusions. It is also worth noting that all analyses from the WoS dataset used the same keyword approach and publication exclusion method as the Scopus data. In particular, this empirical data is supported by a solid theoretical foundation. "Flow" is defined according to the "optimal experience" approach, characterized by deep immersion, enjoyment, and changes in the perception of time. These characteristics make it an extremely relevant and important concept in experience-intensive fields such as tourism. The common and overwhelming trend consistently noted in both databases is the growth of the research field. The main driver of this growth is the strong resonance between the well-established theory of "flow experience" and the disruptive technology of "virtual reality". However, the most noticeable difference is that the analysis of the WoS dataset provides deeper insight into the history and classic works of the field. It emphasizes foundational concepts such as "satisfaction" and can capture early technology papers.

5.2. RQ2: the intellectual bedrock: pinpointing influential voices and venues

Examining core authors, publications, and journals enables us to chart the knowledge landscape of the field. The analysis shows a distinctive move away from psychological theoretical underpinnings toward technology-driven applications within the framework of modern tourism.

The inaugural pillar of Csikszentmihalyi's undying heritage is verified by co-citation analysis in Fig. 7. The co-citation map irrefutably demonstrates that Csikszentmihalyi M., is the intellectual pillar of the entire field. The node representing him, with his name prominently displayed in the network, is significantly larger and centrally positioned, indicating that most research on flow in tourism is built upon his foundational theoretical work on optimal experience. His presence is not only historical; it is a positive and continuous point of reference, demonstrating the deep roots of this field in positive psychology.

The author analysis in Table 5 shows that Albayrak Tahir, Huang Xiaoting, and Liu Chun are the most productive researchers in the dataset, with a focus on technology-mediated experiences such as VR, live streaming, and mobile applications. Meanwhile, an analysis of the most cited articles in Table 7 reveals a clear shift. While previous seminal articles are still pertinent, the most cited recent article is that of Kim and Hall (2019), whose very high citation frequency (TC per Year: 51.71) confirms that the relationship between VR tourism and hedonic motivation (one of the main components of flow) is the most current topic, pushing the knowledge agenda in the field. This project is a significant step, warranting the application of flow-based principles to the new and rapidly evolving field of VR tourism. The study of journals in Fig. 2 shows that leading tourism journals, such as the Asia Pacific Journal of Tourism Research and the Journal of Vacation Marketing, are major platforms for discussion and ideas exchange. However, the study by Kim and Hall (2019), published in the International Journal of Information Management, is particularly significant for its major impact on the field of flow optimization in tourism. This shows the interdisciplinary nature of the field. Flow tourism research is not limited to mainstream tourism journals, but also extends to high-end information systems and management journals. Such recognition confirms that such research, though valuable to tourism theory, also provides insight into human-machine interaction and technology management.

The keyword co-occurrence network analysis in Fig. 6 seems to show a clear split into two main clusters, which include (1) the psychological core (the green cluster) made up of foundational concepts that describe the experience itself, like "flow experience", "immersion", "virtual reality", and "presence". This section answers the "what" and "why" of the phenomenon. (2) The management & application core (the red cluster) contains key business outcomes and marketing concepts such as "satisfaction", "loyalty", "behavioral intention", "destination image", and "brand equity". This appears to answer the "what for" question from a practical perspective. The strong and dense link between "flow experience" and "satisfaction" on the visual map underscores that flow's primary function in this literature is as an influential antecedent leading to positive consumer evaluations.

Meanwhile, Fig. 11 shows the keyword co-occurrence network analysis from the WoS database, which also has some noticeable highlights and similarities, with "flow experience" being the main search term and the conceptual core of this research area. Its direct and strong connections to "flow", "tourism", and "enjoyment" establish the basic building blocks of the field. Studies have focused on individual customer experiences. In addition, the documents indexed in WoS (Fig. 11) show a clear focus on vicarious psychological experiences, primarily through the lens of "Tourism Live Streaming" (TLS). These studies mainly used the SOR model to explain behavioral outcomes.

In contrast, the documents analyzed from the Scopus database (Fig. 6) emphasize managed technology experiences, with the prominent theme of "virtual reality". This field favors the use of the Technology Acceptance Model (TAM) as a theoretical framework and shows stronger links to applied fields such as tourism management and marketing. This reflects the difference in focus between real-time, socially mediated experiences and pre-designed immersive environments.

5.3. RQ3: motivation, foundation, and new frontiers

The thematic map analysis results (Figs. 3 and 4 and Table 8) provide a comprehensive and multidimensional picture of the structure and dynamics of development in tourism flow experience research. Instead of just being separate topic clusters, they form a tightly linked knowledge ecosystem, which suggests a field that has matured and is now in a period of significant transformation.

The focus and most significant influence of the entire field lies in the basic theoretical core, with "flow experience" as an irreplaceable pillar of research. The central position and outstanding connectivity of this topic cluster show that it is not just a single research topic but has become the core theoretical lens for all other research branches. Researchers are no longer focused on redefining the concept of flow, but instead use it as a powerful tool to decode key issues in the tourism industry, such as "satisfaction", "perceived value", and critical behavioral intentions such as "revisit intention" and "destination loyalty".

On that theoretical foundation, the field is being strongly promoted by the dynamic themes of the times, reflecting the shift from "what" to "how". The two main driving forces today are "authenticity" and the applications of "tourism" in a digital context. The rise of terms such as "tourism live streaming" and "consumer behavior" shows that research is focusing on applying flow theory to design and optimize digital tourism experiences, an inevitable trend in the industry. Simultaneously, the theme of "authenticity" affirms a more profound need of tourists: they are not only looking for pleasure, but also craving meaningful, authentic experiences that can put them in a state of flow.

In addition to the mainstream flow, the field is also enriched by niche and emerging themes. Niche themes, such as "customer engagement", are being fleshed out in detail in isolation but are being developed further. If they form an organic connection with the theoretical core, they will be important ideas. In the meantime, emerging topics such as the "technology acceptance model" and "cultural tourism" are efforts to expand the boundaries of the discipline. Although still in the early stages and have not yet had a clear impact, their existence shows excellent potential for integrating the flow experience with other interdisciplinary models.

The overall analysis of the topic clusters from the WoS database in

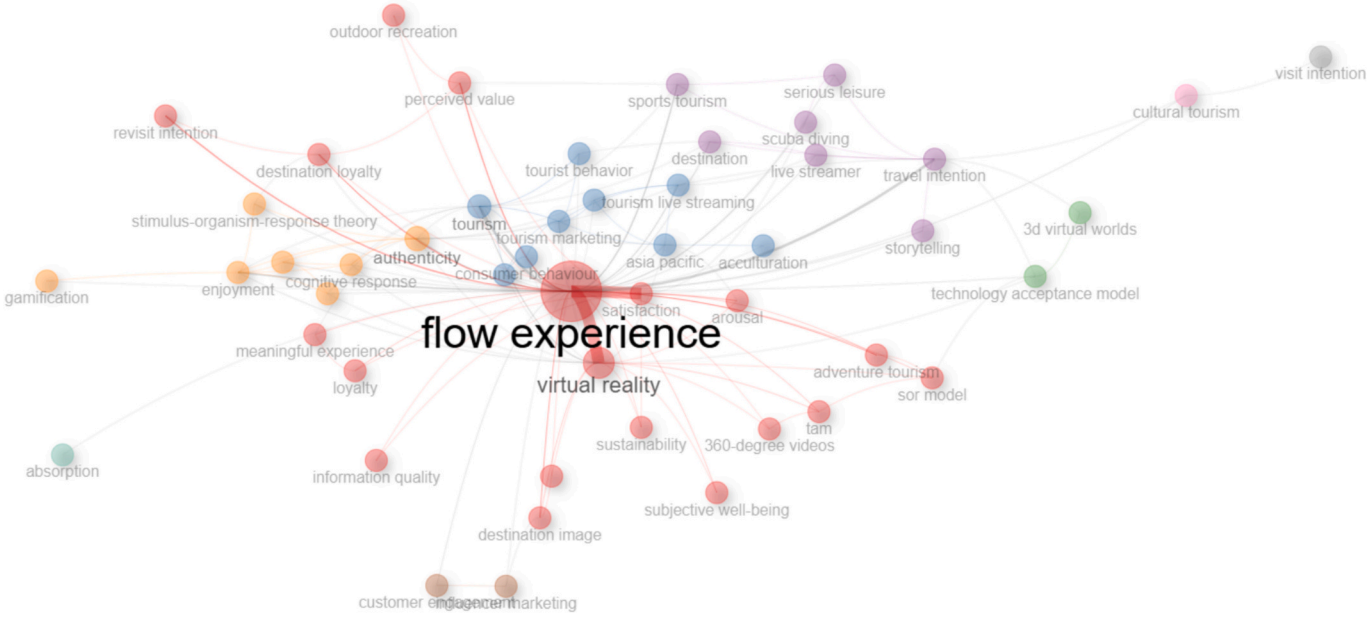


Fig. 4. Co-occurrence network analysis by author keywords. Source: visual map extracted from Bibliometrix.

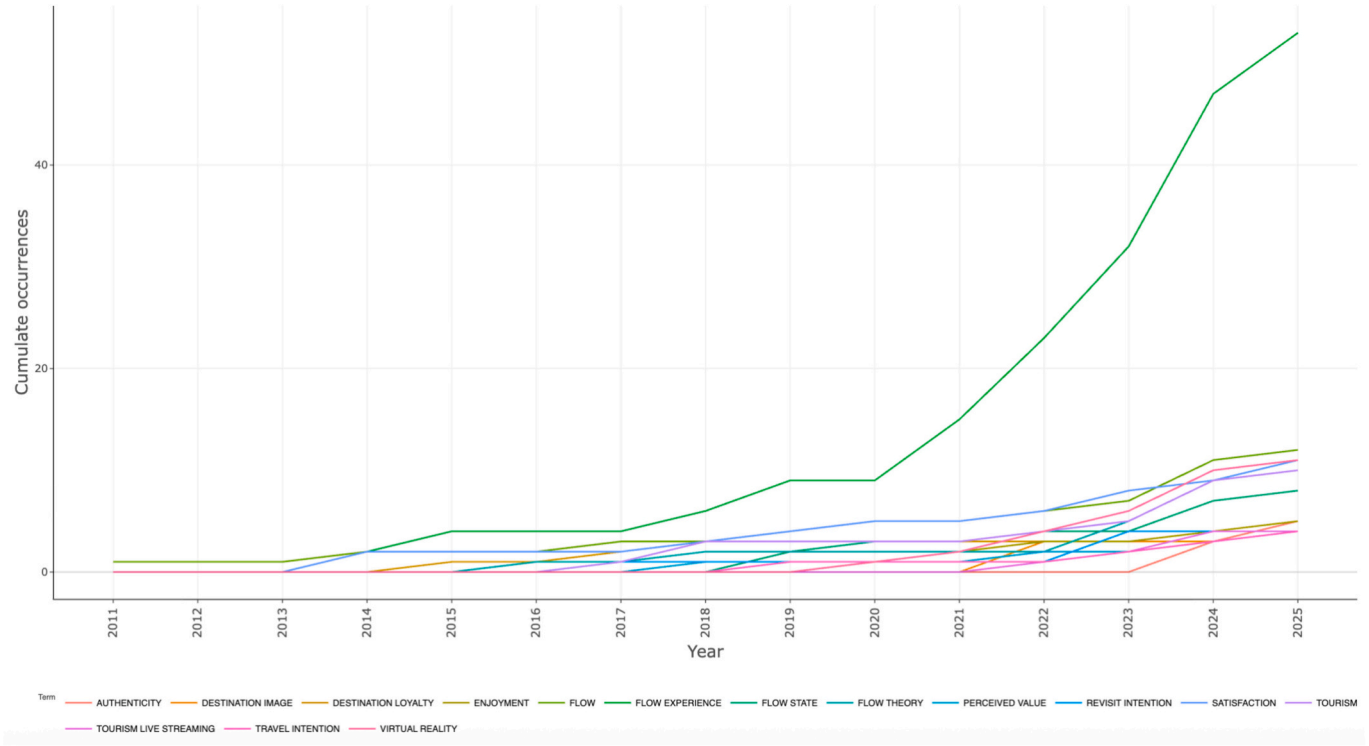


Fig. 5. Frequency of words over time. Source: visual map extracted from Bibliometrix.

Table 9 reveals that "flow experience" remains the core topic, but is secondary in concentration to "authenticity". A lower density indicates a broader, more cross-cutting role and the highest volume of publications. It introduced "satisfaction" as a key topic and showed a more fragmented periphery with specialized topics. The low concentration cluster "behavioral intentions" is weakly linked to the leading network but has a reasonably high density, suggesting some internal coherence. The prominence of "authenticity" and "satisfaction" indicates a focus on the theoretical and psychological foundations of flow experience. The research questions in this collection of materials tend to be "What is the relationship between authenticity and flow experience?" and "Does the flow experience lead to satisfaction?". This is in contrast to the results from Table 9, which focused on the "tourism" application and the behavioral premise of "travel intention".

5.4. RQ4: future research roadmap

Thematic mapping and trend topic analysis suggest that a research pathway has been established through which future research on flow experience in tourism can be conducted. At a theoretical level, the co-

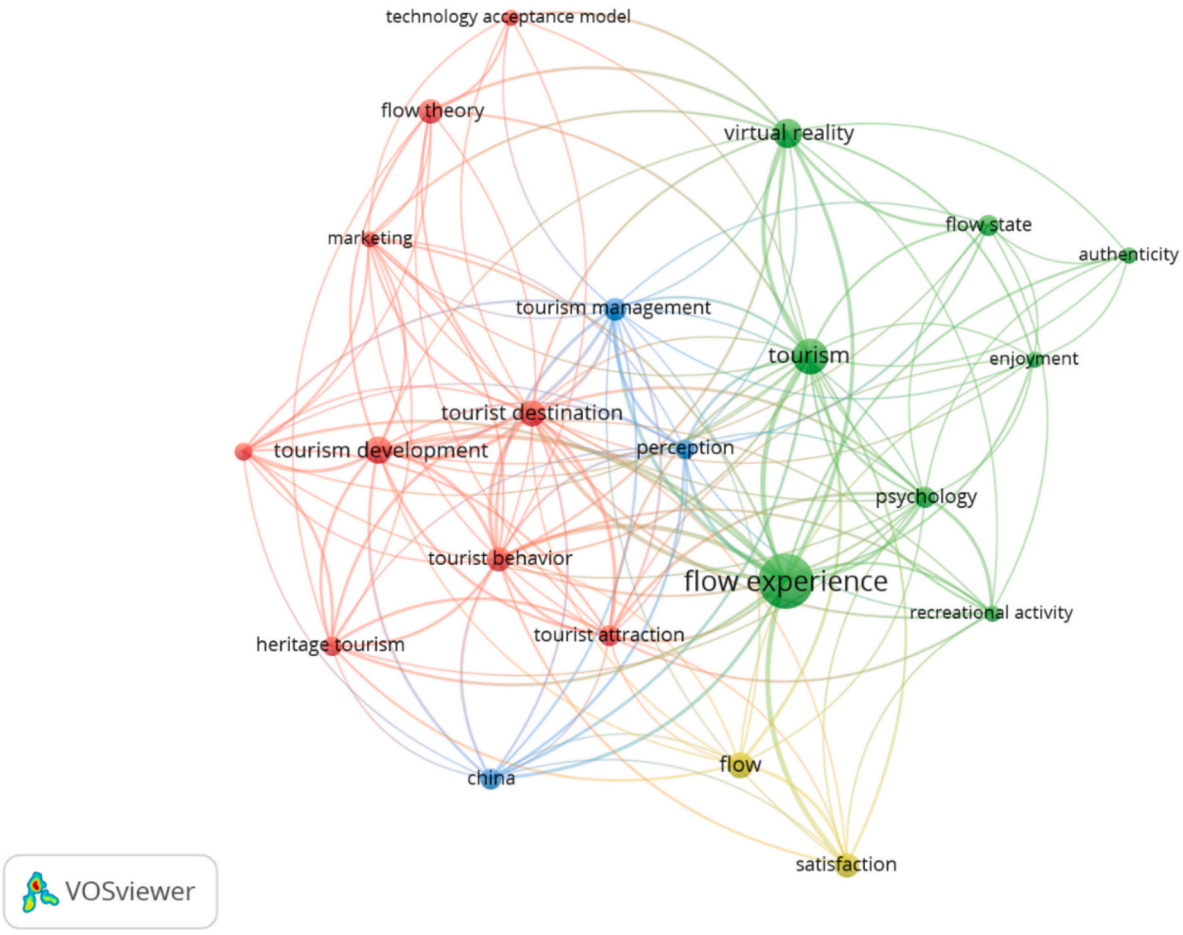


Fig. 6. Keyword occurrence map. Source: visual map extracted from VOSviewer.

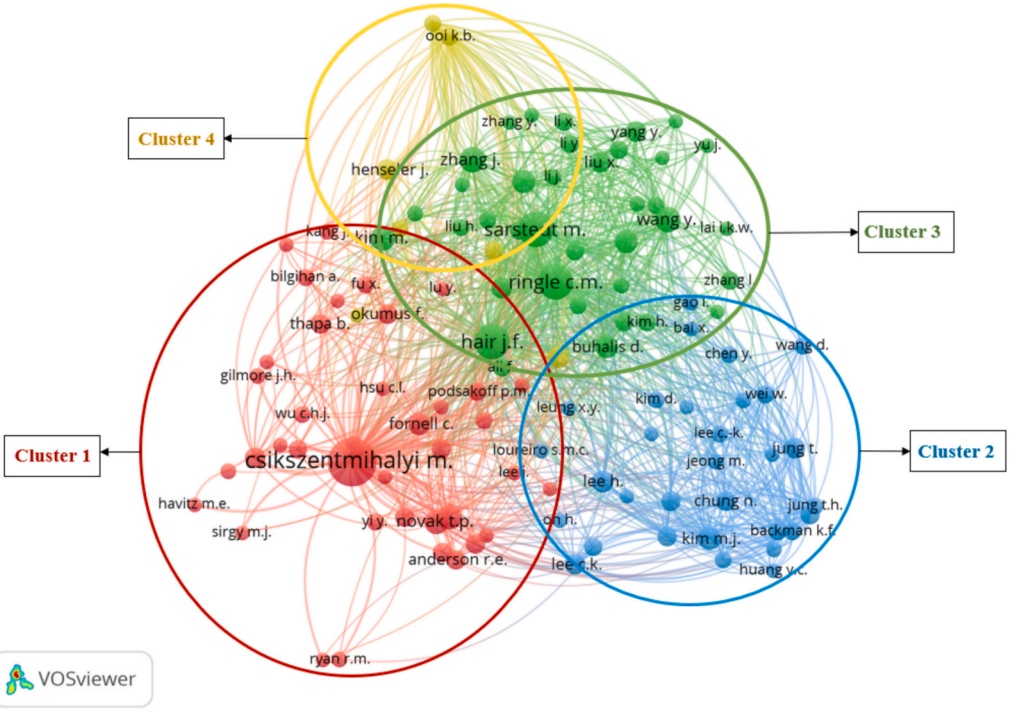


Fig. 7. Co-citation of authors. Source: visual map extracted from VOSviewer.

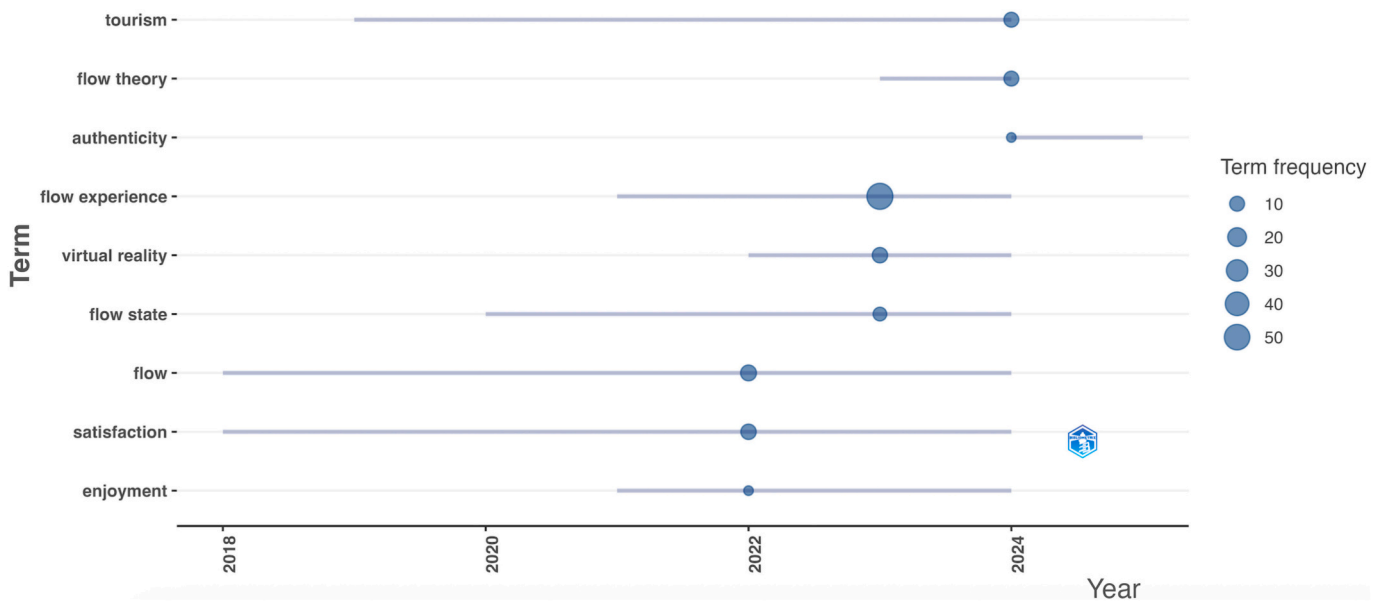


Fig. 8. Trend topics. Source: visual map extracted from Bibliometrix.

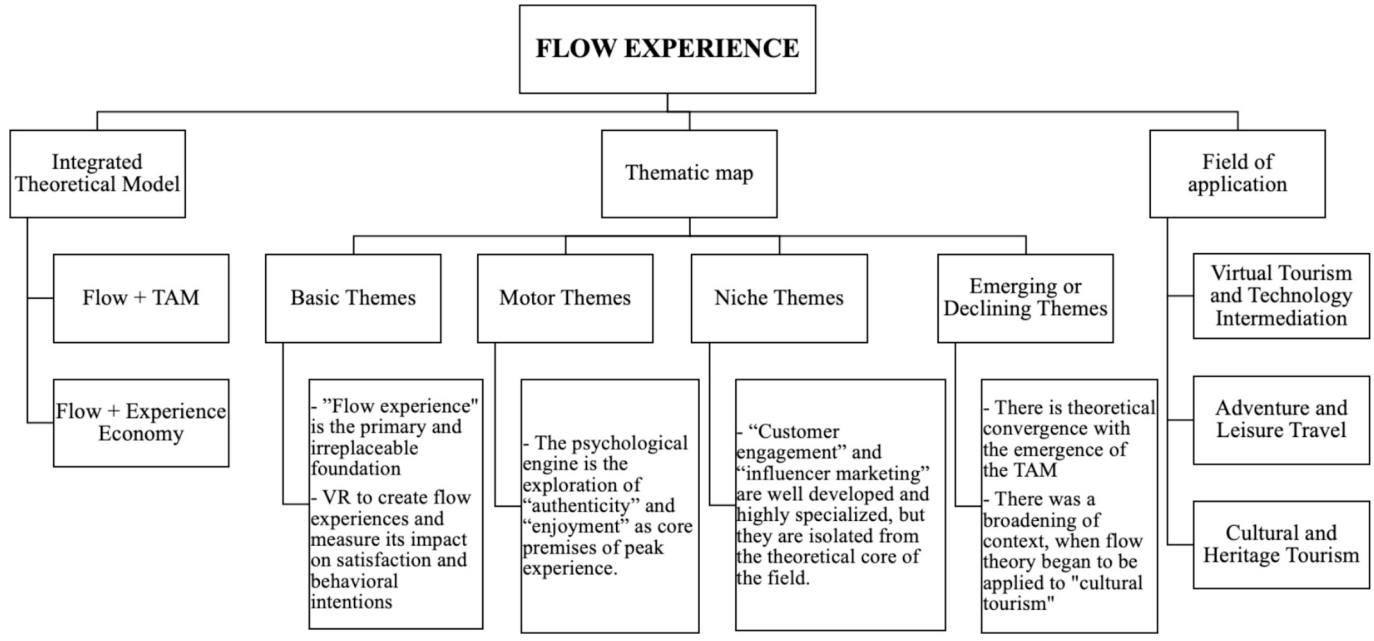


Fig. 9. Insight framework of flow experience in tourism. Source: the author synthesized this from the research results.

occurrence of foundational themes like "flow experience–satisfaction–loyalty" and driving themes like authenticity, live streaming, and virtual reality underlines the need to develop more integrative approaches that connect antecedents, flow experience, and consequences in terms of both the psychological and managerial dimensions of flow. Future research can be informed by the Integrated Flow Experience Framework in Tourism, which can be used as a tool to assess the profile of flow experience, memories, or citizenship consequences resulting from different configurations of antecedents, such as challenge–skill, social interaction, or technological immersion.

At the methodological level, the clusters identified in the thematic map show a strong reliance on cross-sectional survey designs, despite growing interest in dynamic topics such as VR tourism, live-streaming, and serious leisure. Therefore, there is a need for greater emphasis on longitudinal and experimental designs that manipulate concepts such as challenge-skill balance, presence, or interactivity, and that incorporate measures of psychophysiological or behavioral responses to understand the temporal progression of flow experiences better. The use of a bibliometric science-mapping approach, combined with a more qualitative approach such as the critical incident technique or experience sampling, helps establish links between the macro-level research fronts and how tourists report their experiences of flow at various stages of the trip.

At the contextual level, it is clear from the map that the clusters related to Asia-Pacific destinations, adventure and sports tourism, and technology-intensive flow experiences are highly developed, while other flow experience themes, such as wellness and spiritual tourism, cultural and heritage settings, senior and family tourists, and Global South destinations, remain in the periphery and in the emergence stage. Thus, it is suggested in the future research that the flow experience be extended to include these contexts in order to examine other forms of flow

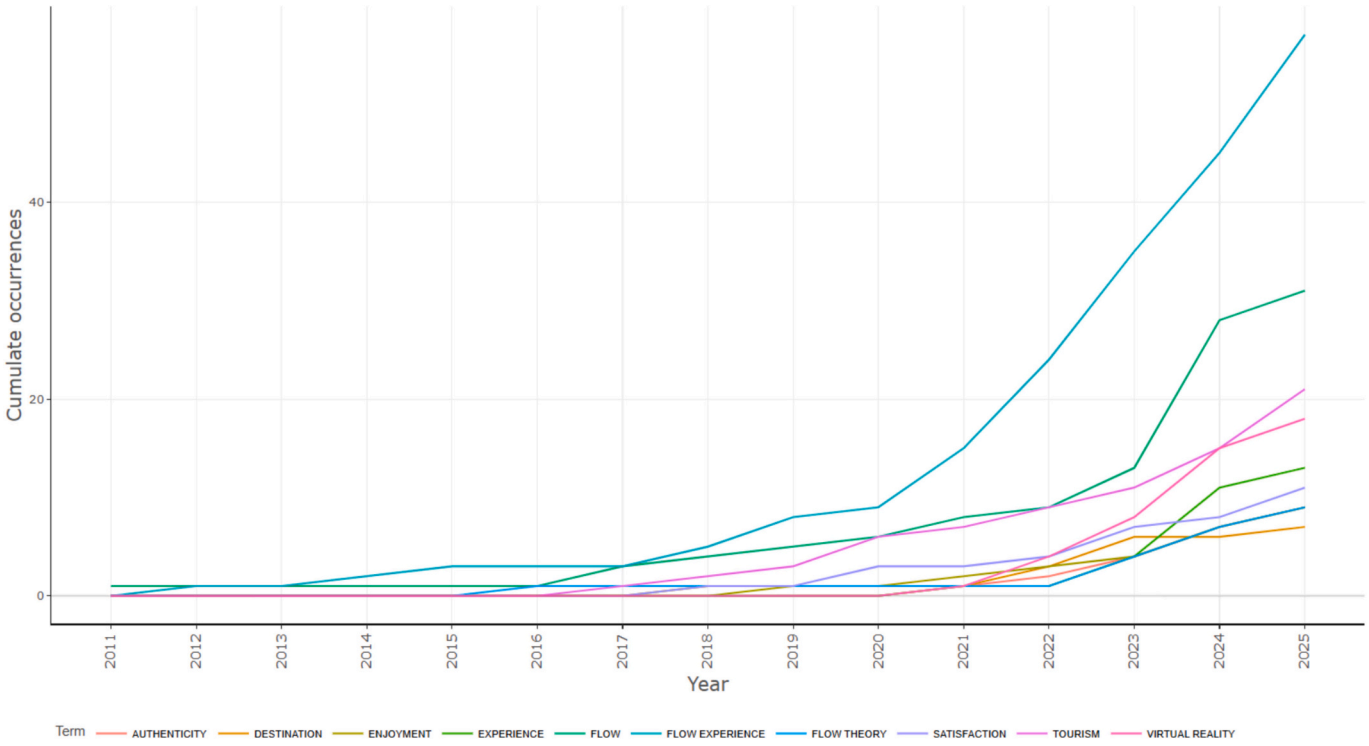


Fig. 10. Frequency of words over time (WOS database). Source: visual map extracted from Bibliometrix.

experience, which are not necessarily related to arousal and might include other forms of flow experience such as contemplation and meaning-making, and test the generalizability of the established relationship between flow experience, satisfaction, and loyalty in other cultural contexts, age groups, and governance systems.

Expanding the research context is an option for future related research. The emergence of TAM as an important topic and the growing body of literature attempting to integrate the two theories point to the most pressing theoretical task. Future research needs to move beyond simply adding flow as a prefix to TAM constructs (perceived usefulness and perceived ease of use). Important research questions include: (1) How does the emotional and experiential nature of flow interact with the cognitive and rational evaluations of TAM? (2) Does flow mediate or moderate the impact of system characteristics on user attitudes? (3) Is it possible to develop a unified "Experiential Technology Acceptance Model" specific to the tourism context?. In addition, methodological innovation is also a worthy approach to capturing the Objective Reality of Flow. Pioneering studies, such as those by Huang et al. (2021) and Liu and Huang (2023), have utilized psychophysiological measures to capture objective, real-time physiological correlates of flow. A pressing appeal for future research is the incorporation of mixed-method designs that combine traditional surveys with objective physiological assessment and qualitative methods in order to create a more detailed and sturdy understanding of the experience of flow.

6. Conclusions

6.1. Theoretical contributions to flow and psychology

First, this research provides a quantitative chronicle of how flow experience research in tourism has evolved between 2011 and 2025, with clear milestones in its intellectual development. From the analysis, there is a clear transition from the use of flow experience as a general psychological concept to its use as a context-specific explanatory tool, especially in technology-based tourism websites, mobile applications, virtual reality, and live streaming.

Second, this study not only lists popular topics by deconstructing themes and identifying the strategic role of each cluster, but also sheds light on how different themes interact and shape the field. The analysis suggests that the field is not developing uniformly but seems to be driven by a few specific intellectual motors, such as "authenticity" and "tourism" (which is pushed by livestreaming), while other areas are still in their infancy ("technology acceptance model") or remain isolated ("customer engagement", "visit intention"). By laying out this thematic map, this study highlights a significant gap and an opportunity for future work to integrate these theoretical streams, for instance, by exploring how a VR flow experience might enhance customer engagement with an influencer.

Third, this research offers the Integrated Flow Experience Framework in Tourism (IFEFT, Fig. 9) as a synthetic meta-framework that structures the field. The IFEFT integrates the existing literature into a new three-pillar construct: antecedents, flow experiences, and consequences, thus rethinking flow as the central Organism (O) in a new S–O–R logic that subsumes the psychological, technological, and managerial into a single antecedent-flow-outcome system. Rather than simply summarizing existing knowledge, this framework organizes the scattered causal relationships (between destination attributes, authenticity, technology immersion, and flow, on the one hand, and satisfaction, loyalty, memories, or well-being, on the other) into a coherent theoretical model that other research in tourism could follow or modify in modeling the effects of new technologies on flow and subsequent behaviors.

Aside from these three major contributions, the review also helps establish the conceptual limits of flow in its broader nomological network. For example, the paper distinguishes flow from mindfulness, engagement, satisfaction, presence, immersion, and peak experience. This helps establish the limits of flow as a fleeting, goal-oriented, optimal experience grounded in a challenge-skill balance and intrinsic enjoyment, rather than positive affect, post hoc evaluation, or a technological property of the medium itself. The results of the science mapping also help theorize the field as comprising a psychological core, centered on flow, authenticity, serious leisure, and well-being, as well as

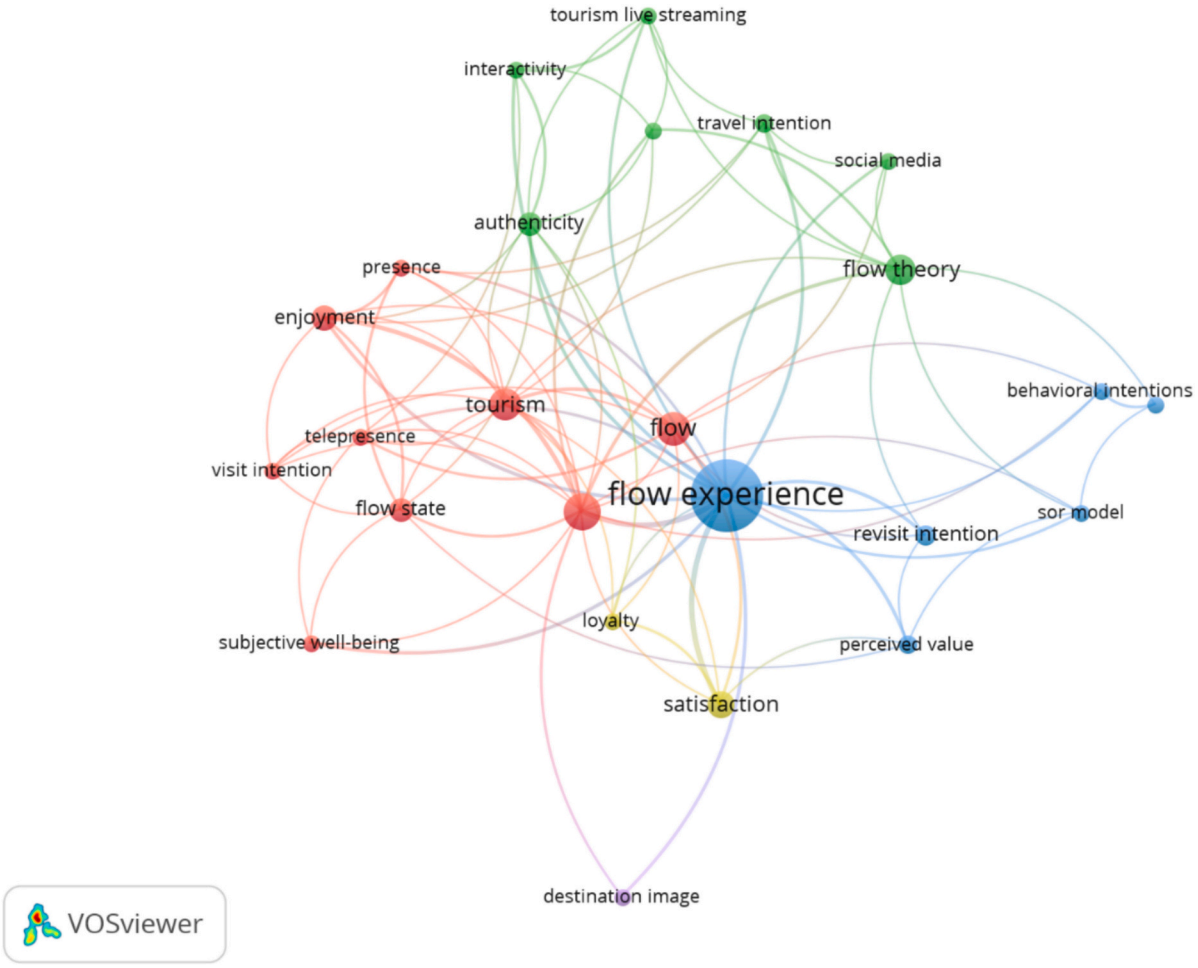


Fig. 11. Keyword occurrence map (WOS database). Source: visual map extracted from Bibliometrix.

Table 9
Quantitative overview of thematic clusters in flow experience research (WOS database). Source: author's synthesis from analysis results on Bibliometrix.

Cluster	Callon's centrality	Callon's density	Cluster frequency
Satisfaction	0.512	36.892	48
Behavioral intentions	0.083	44.792	13
Flow experience	0.664	29.472	180
Cultural tourism	0.000	58.333	7
Authenticity	1.025	39.490	62
Experiences	0.000	50.000	5
Customer engagement	0.000	33.333	3

a managerial/technological core, centered on acceptance and intention models, each employing different explanatory logics on the same construct. The synthesis, therefore, reinforces the place of flow as a central psychological mechanism that converts a wide range of stimuli, whether destination attributes, authenticity, or technology, into a common metric of value, such as satisfaction, loyalty, or well-being, re-orienting the field of tourism experience from a static input-output model to a process-oriented one.

6.2. Practical contributions

Beyond the theoretical contributions, this study also offers some important practical takeaways for managers, marketers, and tourism product developers.

The analysis confirms that immersive technologies, such as VR and AR, are compelling tools for creating flow experiences. What this means in practice goes beyond just using tech as a marketing gimmick. Initially, VR in tourism was primarily viewed as a means to "try before you buy". However, this research suggests that well-designed VR experiences are capable of creating a deep state of flow. This psychological state is valuable and satisfying in its own right for the user. When an experience is valuable in its own right, it can likely be commercialized. Therefore, managers should consider generating revenue directly from virtual tourism experiences that are designed to optimize flow.

This research also taps into the power of authenticity to create flow by identifying "authenticity" as a driving theme with a direct, strong link to the flow experience. This means that tourists are not only seeking authenticity as a destination in itself, but also as a means to encourage immersion and a flow state. The operational implication here is that product development and marketing strategies must truly put authenticity at the forefront. Technology can be used to recreate history authentically, rather than creating a fantasy version, thereby helping tourists achieve a state of flow through a sense of authenticity.

The analysis reveals that "tourism live streaming" is a significant driver of the "tourism" theme, particularly in the Asia-Pacific region. Similarly, the "customer engagement" topic is related to influencer marketing and their ability to create a sense of flow through parasocial interaction. For livestreams, real-time and momentary interactivity needs to be brought to the forefront to create a sense of co-presence and flow among the audience. For influencer marketing, selecting the right influencers who can build deep, trustworthy relationships with their audience is key to converting the viewer's flow experience into actual travel intentions.

6.3. Limitations and future research directions

Although this study provides a thorough overview, it has some limitations that should be noted, which suggest directions for future research. First, this study was based only on the Scopus database and was limited to English-language publications. This work was missed as it was published on other platforms or in other languages, especially Chinese, given the prominence of the Chinese context identified in the analysis. Second, the data for 2025 were only collected up to June, so it likely does not fully reflect the entire year's output in what is a rapidly growing field.

Based on the findings and limitations, this study proposes a structured future research roadmap focusing on three key areas: theoretical integration, methodological innovation, and expanding the research context. Based on the emergence of TAM as a new topic and the dominance of the technological context in flow studies, future studies could focus on addressing the question of how to develop an "Experiential Technology Acceptance Model" (Experiential TAM) that deeply integrates the emotional factors of flow with the cognitive factors of TAM. Additionally, based on the identified theoretical fragmentation between technology-focused and behavior-focused topics, researchers are focusing on how flow mediates or moderates the relationship between system characteristics (such as VR vividness) and behavioral outcomes (such as tour booking intention).

Innovating the methodology will help expand the field of research from the complexity of the flow experience, which requires a combination of the breadth of quantitative data and the depth of qualitative data. What nuances of the flow experience can mixed-methods designs illuminate that single methods cannot capture?

Last but not least, approaching the research topic by broadening the context is equally important. This article has identified less developed research areas, opening up new avenues for exploration. Alternatively, based on defining "authenticity" as a dynamic and pioneering topic, it indicates an emerging and promising field of study.

CRediT authorship contribution statement

Nguyen Binh Phuong Duy: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical compliance

This study did not use survey data or any information related to human participants.

Funding statement

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Declaration of competing interest

The author has no conflicts of interest.

Data availability

The data that support the findings of this study are openly available in Zenodo.

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